

DATA 423

ASSIGNMENT

01

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Table of Contents

1). Introduce Data:	3
2). Missingness Plot:	9
3). Time Series Plot (Part One):	12
4). Rising Order Chart:	15
5). Correlation Plot:	21
6). Time Series Plot (Part Two):	25
7). Boxplot (Part One):	27
8). ggPairs Plot:	30
9). Boxplot (Part Two):	34
10). Time Series Plot (Part Three):	36
11). Tabplot Chart:	37
12). Mosaic Chart:	39
13). Data Table:	39
14). Conclusion:	41
15). Use of AI:	41

Note to Marker(s): Some of the visualisations used in the report do not have the full title in the report and other information. However, they are full visible within the Shiny application and there is user feedback to what selection criteria was used to create the visualisation.

Sensor Groupings:

- **Group A:** *sensor4, sensor8, sensor11, sensor16, sensor22, sensor24, and sensor28.*
- **Group B:** *sensor21, sensor23, sensor25, sensor26, sensor27, sensor29, and sensor30.*
- **Group C:** *sensor12, sensor13, sensor14, sensor15, sensor17, sensor18, sensor19, and sensor20.*
- **Group D:** *sensor1, sensor2, sensor3, sensor5, sensor6, sensor7, sensor9, sensor10, and Y.*
- **Group E:** *sensor1, sensor2, sensor3, sensor4, sensor5, sensor6, sensor7, sensor8, sensor9, sensor10, and Y.*
- **Group F:** *sensor11, sensor12, sensor13, sensor14, sensor15, sensor16, sensor17, sensor18, sensor19, and sensor20.*
- **Group G:** *sensor21, sensor22, sensor23, sensor24, sensor25, sensor26, sensor27, sensor28, sensor29, and sensor30.*

1). Introduce Data:

The dataset used in this exploratory data analysis is named *Ass1Data.csv*, which is an artificially generated dataset with a file size of 221 KB. The dataset contains 360 rows of observations and 44 columns of variables, giving a total of 15,840 cells (rows × columns). Out of these values, 15,319 entries are present, while 521 values are missing. There are 201 rows that are complete, representing 55.8% of the dataset. Overall, the dataset has a data completeness of approximately 96%, meaning the majority of the dataset contains recorded values.

Within the dataset, there are 31 numeric variables (70.5%), with 30 of these representing sensor data measurements (named *sensor1* to *sensor30*). The other remaining numeric variable is labelled *Y*, which appears to be the dependent variable of interest. A statistical summary of these variables can be found in **Table 01**. All sensors contain between 1.9% and 5% missing values except for *sensor6*, which contains 28.9% missing observations. Variable *Y* is complete and has no missing values.

In addition to the numeric variables, there are 12 categorical variables. Of these categorical variables, five have an inherent ordering. These ordered factor variables include *Priority*, *Duration*, *Temp*, *Price*, and *Speed*, where each category represents an increasing level (for example low to high or slow to fast). These variables are summarised in **Table 02**. The remaining categorical variables are unordered factors, including *ID*, *Operator*, *Location*, *Agreed*, *State*, *Class*, and *Surface*, which are summarised in **Table 03**.

Examination of the categorical variable distributions reveals several notable imbalances that warrant attention. Among the ordered factors in **Table 04**, *Price* is heavily skewed toward *Expensive* (54.72%), while *Fair* represents only 10.83% of observations, suggesting the dataset captures predominantly higher-value transactions. Similarly, *Temp* shows a bias toward warmer conditions, with *Cold* comprising 8.33% of recordings. *Duration* has *Short* (44.72%) dominating over *Very Long* (15.56%), indicating most operations conclude relatively quickly.

The non-ordered factors in **Table 05** exhibit further disparities. *Operator* is overwhelmingly dominated by *IK* at 60.28%, while *OL* only has 7.50%. *Class* reveals *Wooden* (42.22%) as the primary material and *Metal* appearing in fewer than 9% of observations. These distributional imbalances will be explored further within the Tabplot and Mosaic Plot to determine whether they reflect genuine operational patterns or potential sampling biases that could influence subsequent analyses.

The *ID* variable is used as the main identifier within the dataset. A manual check showed that each observation has a unique identifier and there were no missing values within that variable. The sequence of IDs appears to follow a mostly ordered pattern, beginning with values ranging from *G9937* to *G9948*, then switching to a different prefix with *D13* to *D75*, before continuing again with *G10012* through to *G10296*. In total, there are 63 observations with a *D* prefix (17.5%) and 297 observations with a *G* prefix (82.5%).

The dataset also contains a *Date* variable, with observations ranging from 12/01/2018 to 28/11/2024, covering a period of 2,512 days. Across these years there are approximately 51 to 53 observations recorded per year, with the exception of 2024, which contains 48 observations.

Each observation appears to have been recorded approximately one week apart from the previous one in a natural chronological order. A manual check of the dates shows that observations consistently occur on either a Thursday or a Friday, suggesting a regular weekly recording schedule. Of the total observations, 179 (49.7%) occur on a Thursday and 181 (50.3%) occur on a Friday,

indicating an almost even distribution between these two weekdays. All recorded dates were valid and followed this weekly trend.

The dataset structure therefore consists of the response variable Y , an identifier variable ID , several operational categorical variables, and 30 numeric sensor variables. These sensor variables represent measured readings with varying value ranges and some missing observations. Most sensors contain around 340–353 recorded values, although *sensor6* contains fewer recorded values (256 observations), indicating a higher level of missing data compared to all other variables.

Overall, the dataset represents a structured collection of sensor readings, operational conditions, and categorical descriptors recorded on a weekly basis, which appear intended for analysing their relationship with the response variable Y . These variables will be explored further in the subsequent sections through statistical summaries and visualisations.

Table 01: Numeric Variables:

Variable	Min	Q1	Median	Mean	Q3	Max	SD	Missing Count	Missing %
Y	-5.71	14.85	20.17	19.88	25.76	39.39	7.98	0	0%
sensor1	-13.84	9.75	20.71	20.48	31.46	48.43	13.16	11	3.10%
sensor2	-11.37	9.51	19.79	20.15	30.44	49.95	13.27	10	2.80%
sensor3	-12.48	9.68	20.39	20.13	29.45	46.33	12.73	10	2.80%
sensor4	-5.16	11.13	25.54	78.55	38.08	369.62	125.02	12	3.30%
sensor5	-8.39	10.18	20.33	20.24	31.07	46.57	12.93	15	4.20%
sensor6	-9.1	7.96	14.14	19.92	33.13	47.82	14.78	104	28.90%
sensor7	-10.87	9.29	19.92	19.83	30.23	50.3	12.56	7	1.90%
sensor8	-9.61	12.31	23.77	77.9	38.61	367.8	125.35	9	2.50%
sensor9	-8.54	9.07	19.84	19.66	30.31	45.35	12.66	13	3.60%
sensor10	-11.3	10.65	21.14	20.41	30.19	46.32	12.52	12	3.30%
sensor11	0.08	49.56	69.9	113.93	95.8	368.12	110.04	10	2.80%
sensor12	13.63	46.96	64.03	64.85	83.05	120.84	22.83	7	1.90%
sensor13	10.58	47.29	64.74	65.4	83.4	116.05	22.51	11	3.10%
sensor14	19.08	45.78	64.54	64.53	82.48	119.84	22.26	7	1.90%
sensor15	16.38	46.92	63.5	64.16	82.92	119.87	21.79	11	3.10%
sensor16	16.47	51.29	71.61	115.69	94.16	369.09	110.77	16	4.40%
sensor17	11.77	48.19	64.16	65.09	84.43	116.27	22.25	9	2.50%
sensor18	14.09	46.41	62.45	64.08	82.11	110.51	22.25	7	1.90%
sensor19	1.69	48.01	65.56	64.82	83.64	114	22.77	16	4.40%
sensor20	9.07	45.93	63.69	64.74	83.75	119.02	22.66	10	2.80%
sensor21	-77.22	-16.87	21.02	22.69	62.94	125.18	45.34	18	5%
sensor22	-56.24	-8.85	39.8	81.63	83.36	369.16	131.1	11	3.10%
sensor23	-69.07	-12.49	24.42	24.05	62.53	119.09	43.39	14	3.90%
sensor24	-62.88	-6.11	39.96	81.4	88.46	368.91	130.32	13	3.60%
sensor25	-71.84	-14.75	18.76	21.83	59.08	119.51	44.22	9	2.50%
sensor26	-63.56	-12.16	24.57	24.38	59.75	146.29	44.95	13	3.60%
sensor27	-88.91	-15.78	20.63	22.48	60.78	118.16	45.22	8	2.20%
sensor28	-72.43	-8.66	39.58	80.79	84.57	369.52	131.6	11	3.10%
sensor29	-69.84	-12.77	20.64	23.83	62.62	117.02	44.54	12	3.30%
sensor30	-70.51	-13.01	21.28	23.93	60	123.98	43.65	12	3.30%

Table 01: Summarises the key statistical characteristics of the 31 numeric variables. All sensor variables have similar levels of missing data, generally between 1.9% and 5% except for *sensor6* having substantially more missing values at 28.90%. The sensors also display varying value ranges and standard deviations, indicating differences in measurement variability. Several sensors exhibit extreme maximum values or large negative readings, suggesting the presence of distinct value ranges across some sensors. These patterns may indicate potential groupings of sensors with similar measurement behaviour. These distributional patterns will become clearer when examining the visualisations presented in the following sections. Outside of the sensor variables, the dataset contains *Y* as the remaining numeric variable which does not contain any missing values.

Table 02: Ordered Factor Variables:			
Variable	Type	Levels	Values
Priority	Ordered Factor	3	Low, Medium, High
Price	Ordered Factor	3	Cheap, Fair, Expensive
Speed	Ordered Factor	3	Slow, Medium, Fast
Duration	Ordered Factor	3	Short, Long, Very Long
Temp	Ordered Factor	3	Cold, Warm, Hot

Table 02: Summarises the ordered categorical variables in the dataset. These variables have a natural ranking among their levels. The ordered factors include *Priority*, *Price*, *Speed*, *Duration*, and *Temp*, each with three levels (for example, *Priority* ranges from Low to High). These variables represent ranked operational conditions that may be relevant for later analysis.

Table 03: Factor Variables (without an order):			
Variable	Type	Levels	Example Values
ID	Factor	360	G9937, D13, G10296 ...
Operator	Factor	4	HN, IK, OL, TX
Location	Factor	3	Control room, Steam plant, Storage tanks
Agreed	Factor	2	No, Yes
State	Factor	3	Checked, Uncertain, Under review
Class	Factor	4	Ceramic, Metal, Plastic, Wooden
Surface	Factor	3	Rough, Smooth, Textured

Table 03: Summarises the unordered categorical variables in the dataset. Unlike ordered factors, these variables do not have a natural ranking between their levels. The factor variables include *ID*, *Operator*, *Location*, *Agreed*, *State*, *Class*, and *Surface*, each capturing different categorical attributes associated with the observations. The *ID* variable serves as a unique identifier for each record, while the remaining variables describe qualitative characteristics such as operator assignment, measurement location, review status, material classification, and surface type. Collectively, these factors provide contextual categorical information that complements the numeric sensor measurements.

Table 04: Ordered Factor Variables Breakdown:		
Variable:	Value:	Percentage:
Priority	Low	28.33%
	Medium	41.11%
	High	30.56%
Price	Cheap	30.56%
	Fair	10.83%
	Expensive	54.72%
	NA	3.89%
Speed	Slow	9.72%
	Medium	43.33%
	Fast	42.50%
	NA	4.44%
Duration	Short	44.72%
	Long	35.56%
	Very Long	15.56%
	NA	4.17%
Temp	Cold	8.33%
	Warm	50.83%
	Hot	38.06%
	NA	2.78%

Table 04: The ordered categorical variables reveal distinct distributional patterns across the dataset. *Price* is heavily concentrated in the *Expensive* category (54.72%), with *Cheap* (30.56%) as a secondary cluster and *Fair* substantially underrepresented (10.83%). *Duration* shows a decreasing frequency as length increases, with *Short* (44.72%) being the most common, followed by *Long* (35.56%), and *Very Long* (15.56%). *Temp* is predominantly *Warm* (50.83%) and *Hot* (38.06%), while *Cold* observations are scarce (8.33%), indicating a warm bias in data collection. *Speed* is relatively balanced between *Medium* (43.33%) and *Fast* (42.50%), with *Slow* (9.72%) being notably rare. *Priority* displays moderate spread, with *Medium* (41.11%) leading, followed by *High* (30.56%) and *Low* (28.33%). NA rates remain consistently low across all variables (under 4.5%), indicating reliable data collection practices.

Table 05: Non-Ordered Factor Variables Breakdown

Variable:	Value:	Percentage:
ID Prefix	D-prefix	17.50%
	G-prefix	82.50%
Operator	IK	60.28%
	TX	17.50%
	HN	14.72%
	OL	7.50%
Location	Steam plant	35.56%
	Storage tanks	32.50%
	Control room	30.00%
	NA	1.94%
Agreed	No	48.06%
	Yes	51.94%
State	Checked	38.33%
	Under review	35.00%
	Uncertain	24.44%
	NA	2.22%
Class	Wooden	42.22%
	Ceramic	28.61%
	Plastic	16.67%
	Metal	8.89%
	NA	3.61%
Surface	Rough	41.94%
	Textured	30.83%
	Smooth	24.44%
	NA	2.78%

Table 05: The non-ordered variables highlight imbalances similar to the ordered variables. *Operator* exhibits extreme concentration, with *IK* representing over 60% of all observations, while *TX* (17.50%), *HN* (14.72%), and *OL* (7.50%) are underrepresented. *Location* achieves balance across *Steam plant* (35.56%), *Storage tanks* (32.50%), and *Control room* (30.00%), providing balanced spatial representation. *Agreed* approaches near perfect balance at 51.94% *Yes* and 48.06% *No*. *Class* shows *Wooden* (42.22%) as dominant, followed by *Ceramic* (28.61%) and *Plastic* (16.67%), while *Metal* (8.89%) is notably scarce. *Surface* textures are well-distributed, with *Rough* (41.94%) most prevalent, then *Textured* (30.83%) and *Smooth* (24.44%). *State* categories are relatively balanced between *Checked* (38.33%), *Under review* (35.00%), and *Uncertain* (24.44%). NA rates remain consistently low across all variables (under 4%), indicating reliable data collection practices.

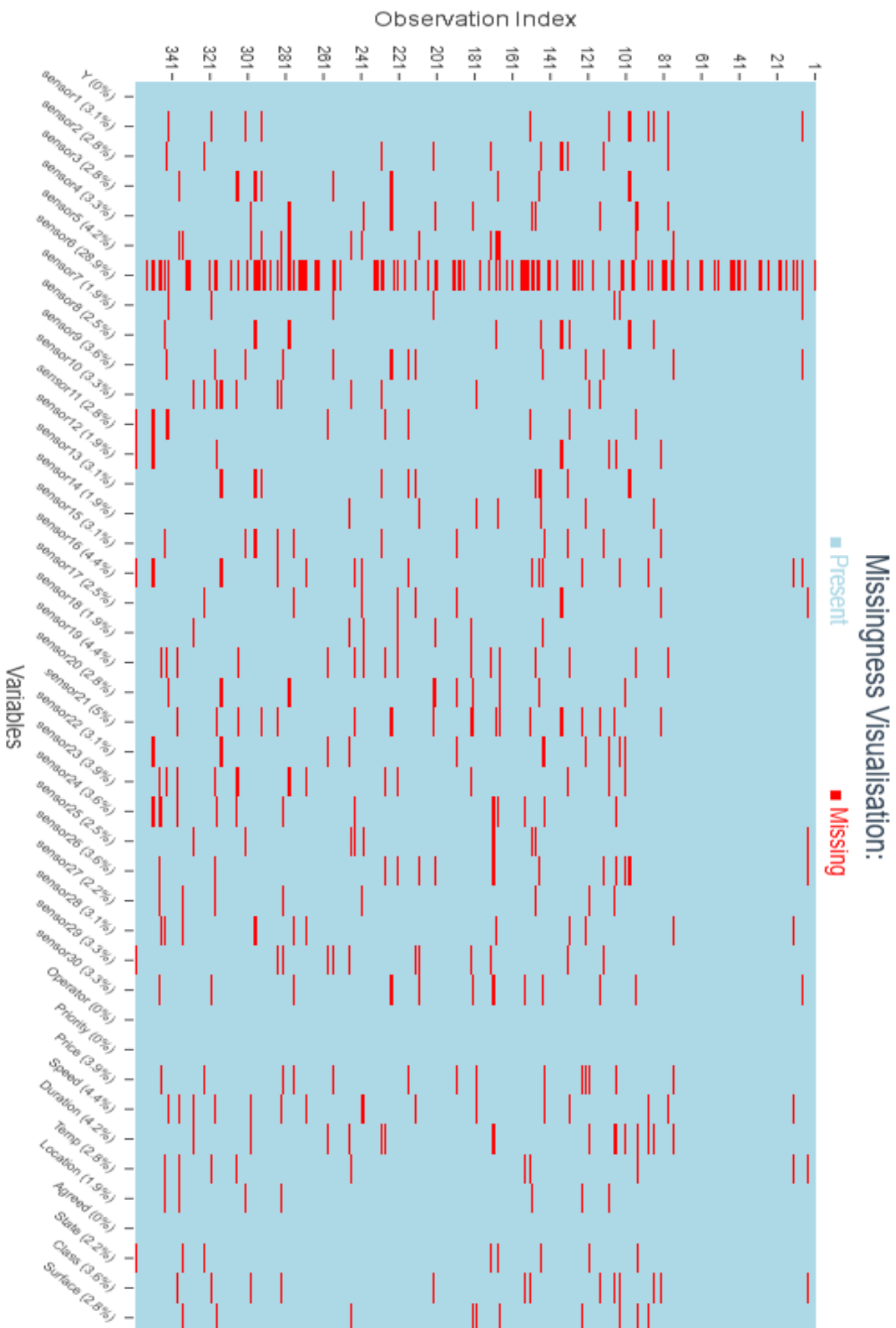
2). Missingness Plot:

A missingness plot was produced to visually examine the completeness of the dataset. This visualisation highlights missing values and allows patterns of missing data to be identified across both variables and observations. As described previously, the dataset contains 15,840 total cells, of which 15,319 entries are non-missing, giving an overall data completeness of approximately 96%, with 521 values recorded as missing.

In the visualisation, missing values are represented in red, while non-missing values appear in light blue. Each column corresponds to a variable and displays its proportion of missing values, while each row represents an individual observation. The dataset is ordered chronologically, covering the period 12/01/2018 to 28/11/2024, meaning the rows reflect the full temporal order of data collection. The Date and ID variables were manually checked for completeness and uniqueness and were found to contain no missing values. As a result, they were removed from the missingness visualisations to improve clarity. **Visualisation 01** gives a good complete overview of the overall missingness of the dataset.

The most notable feature of the missingness plot is variable *sensor6*, which contains substantially more missing values than any other variable in the dataset. *Sensor6* has 104 missing observations out of 360 (28.9% missingness), which is nearly six times higher than the next highest variable, *sensor21*, which contains approximately 5% missing values. Apart from this variable, most other sensors have either no or relatively low and consistent levels of missing data. When examining the pattern of the missingness of *sensor6*, there was no clear clustering or systematic pattern across observations or variables. Instead, the missing values appear relatively scattered throughout the dataset. This suggests that the missingness may not be strongly associated with specific time periods or combinations of variables. The unusually high level of missingness in *sensor6* is therefore a noteworthy finding and may require further consideration if used for further analysis or modelling work. **Visualisation 02** displays the complete missingness of that specific variable.

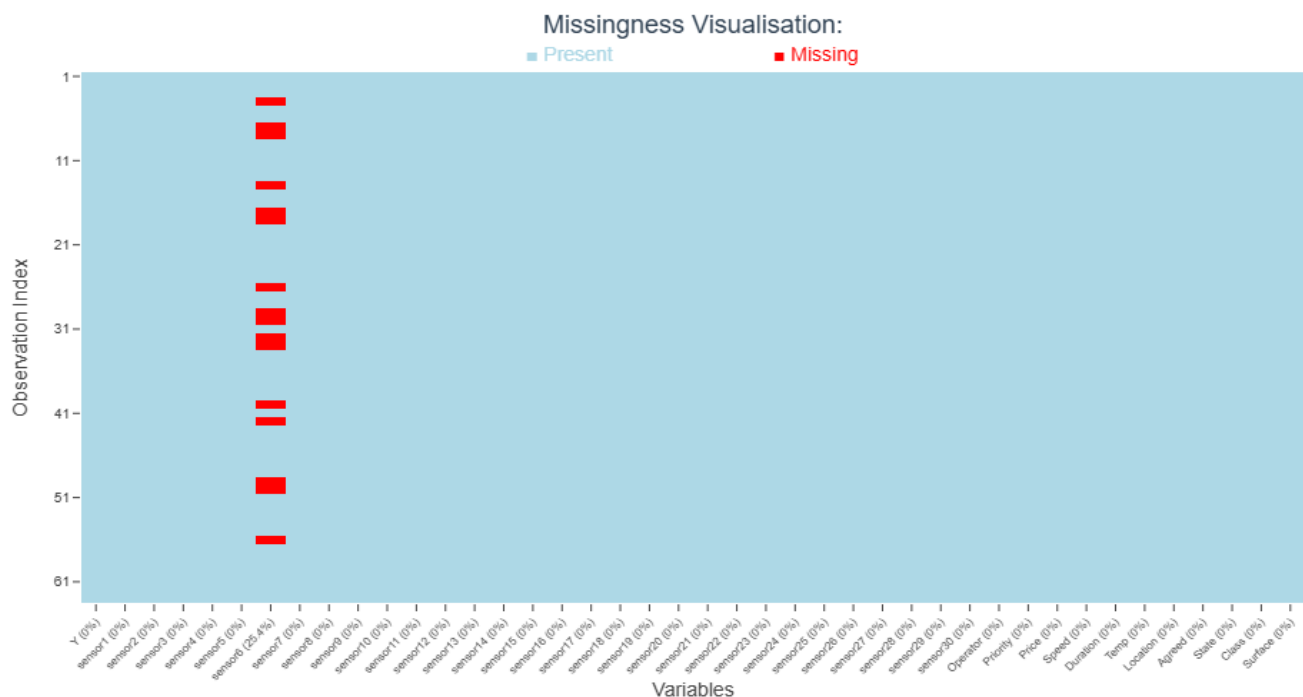
Another notable feature observed in the missingness plot is a continuous period with no missing values across the dataset except for *sensor6*. This period occurs between 31/03/2018 and 20/06/2019. During this time frame, 63 observations were recorded, and only 16 missing values occurred, all of which belonged to *sensor6*. This indicates that during this period, all other variables were fully recorded, suggesting a phase of particularly complete data collection. The reason for this unusually complete period is not immediately clear using the unfiltered missingness chart alone and warrants further investigation. **Visualisation 03** best demonstrates this period of non-missing values.



Visualisation 01: A missingness plot displaying all variables in the dataset across the full observation period of 12/01/2018 to 28/11/2024. Red cells indicate missing values while light blue cells represent recorded values. Each column corresponds to a variable, and each row represents an observation ordered chronologically. There were 521 missing values found to be missing out of 15,840 total cells. This means the dataset was 96% complete.



Visualisation 02: Complete missingness pattern of *sensor6*. There are 104 missing observations out of 360 (28.9% missingness). The pattern of the missingness does not appear to have a noticeably obvious connection to other variables or is it chronologically based. Further investigation to this sensor in terms of a data quality issue is warranted and addressed if desired to be used in future analysis or modelling work.



Visualisation 03: Complete missingness pattern of all variables from 31/3/2018 to 20/6/2019. There are 63 observations recorded during this period with only 16 missing values. All missing values came from *sensor6*. This means that all other variables did not have a non-complete entry during this period and is uncharacteristic of the rest of the dataset. This characteristic about the completeness of this time period warrants a further investigation. Please bear in mind that the observation index in this visualisation is filtered and actually goes down from observation 13 to 75 within the dataset instead of 1 to 62 as labelled on the visualisation.

3). Time Series Plot (Part One):

A time series plot is an effective way to capture temporal trends in numerical data. All numeric variables were plotted on a time series graph and compared with one another over the period from 1st December 2018 to 28th November 2024. **Visualisation 04** displays the raw values across this timeframe showing a generally stable range with a distinct cluster of high values breaking away in the early period.

Although the graph appears cluttered, a distinct group of variables becomes noticeable, breaking away from the main cluster between mid-2018 and mid-2019. Further investigation revealed that the sensors involved in this pattern are *sensor4*, *sensor8*, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28*, which correspond to the notable period identified in **Visualisation 03** (31st March 2018 to 20th June 2019). These seven sensors were then plotted separately in **Visualisation 05** across the full timeframe from 1st December 2018 to 28th November 2024, without any filters applied.

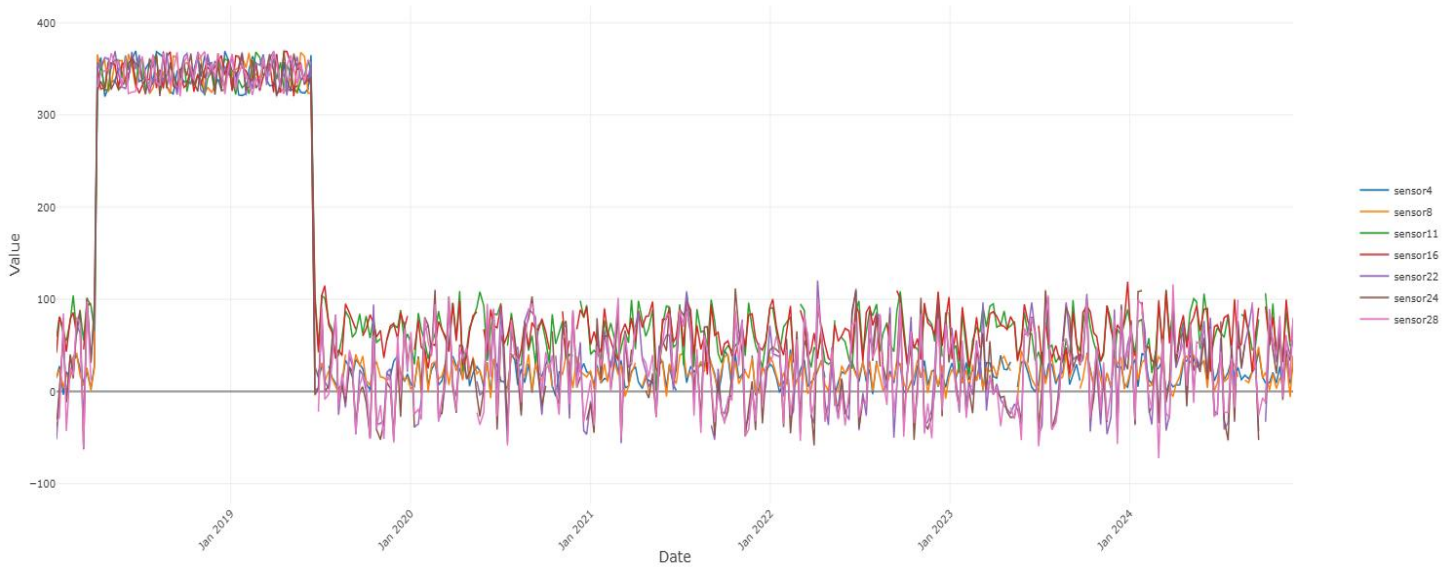
A time series analysis of variable *sensor6* was conducted. As discovered within **Visualisation 02**, **Visualisation 06** displays the time series trend of the sensor and the degree of missingness present. From the trend presented, there is no obvious discernible pattern to how the missingness values have come about from the values shown. The visualisation was plotted from 1 December 2018 to 28 November 2024.

Time Series Plot



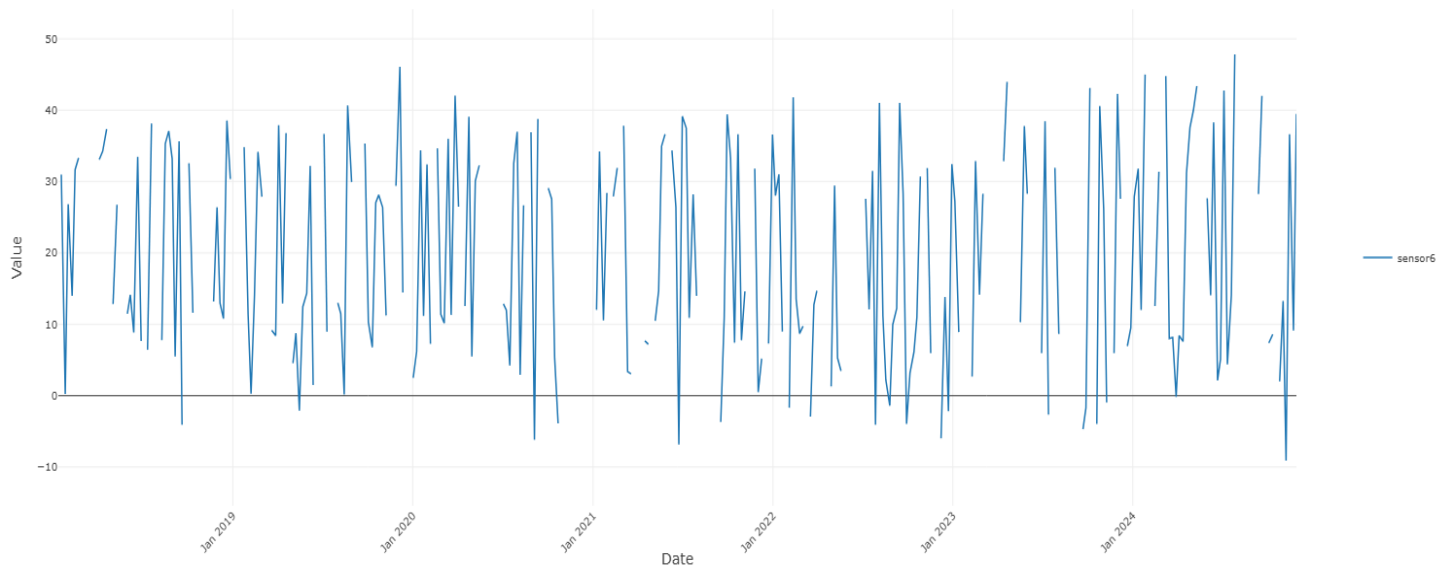
Visualisation 04: Displays the raw values of all numeric variables across 1st December 2018 to 28th November 2024, without any filters. Although the graph appears cluttered, a distinct group of variables becomes noticeable, breaking away from the main cluster between mid-2018 and mid-2019.

Time Series Plot



Visualisation 05: Displays the raw values of the seven sensors that broke away from all other numeric readings from Visualisation 04. The seven sensors that contributed to these extreme large values during the period of 31st March 2018 to 20th June 2019 were *sensor4*, *sensor8*, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28*. The values of these sensors during that period clustered between 300 to 400 units throughout that period before dropping down to between -50 and around 100 units. This time period lines up with the same period that had no missing variables identified as seen in Visualisation 03. Further investigation into these sensors grouping and to why they group together for that period is warranted.

Time Series Plot



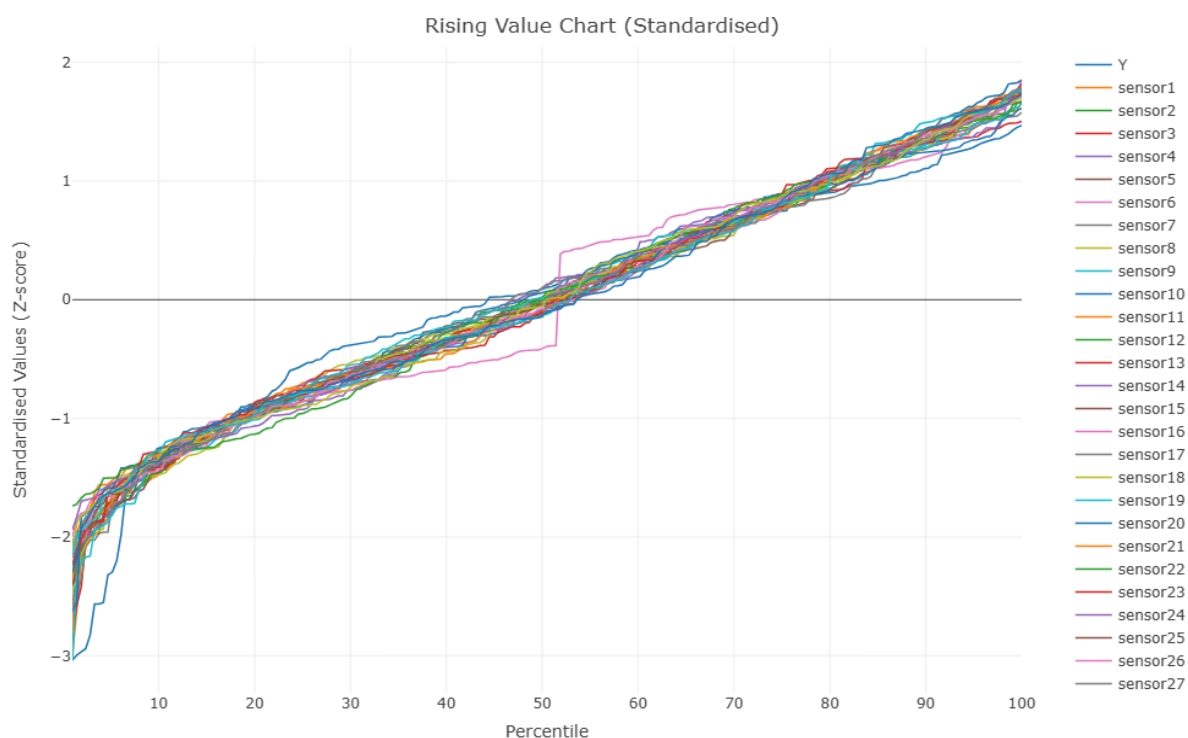
Visualisation 06: Displays the time series trend of *sensor6* from 1st December 2018 to 28th November 2024, highlighting the degree of missingness present. As observed, there is no obvious discernible pattern to explain how the missingness values have arisen from the data shown.

4). Rising Order Chart:

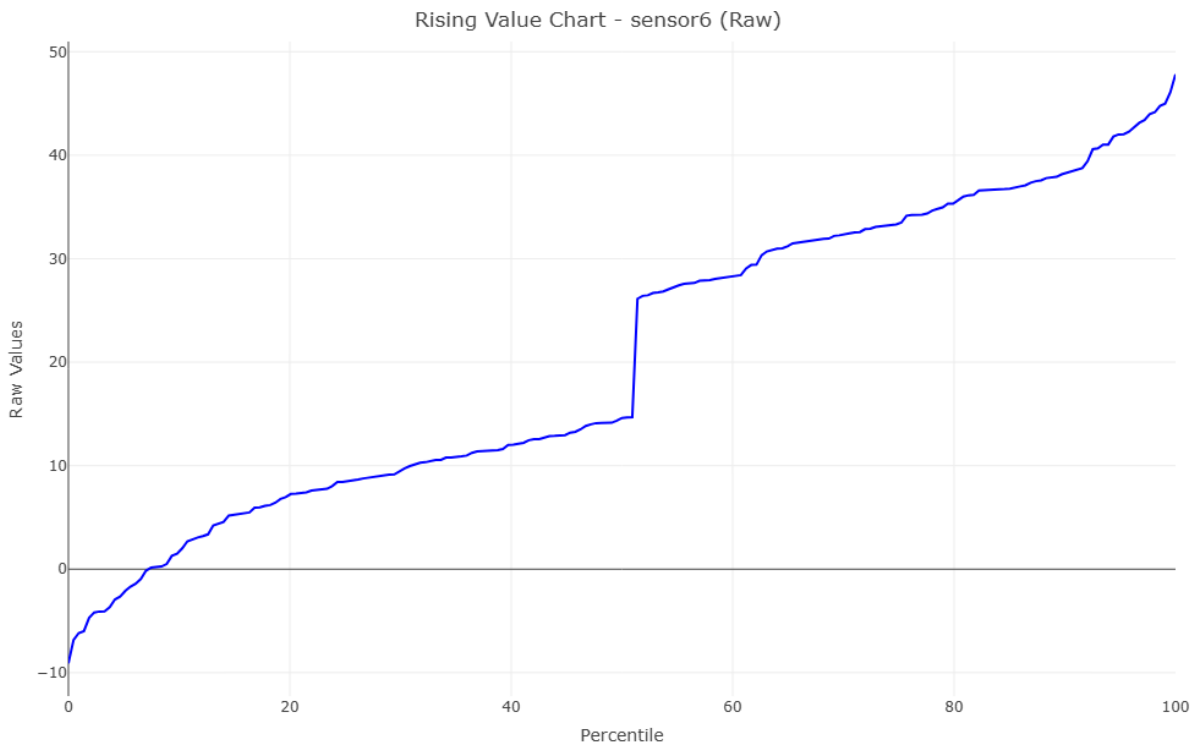
A rising order chart is a visualisation used to explore how the values of a variable increase across the dataset. It is created by arranging the observations in ascending numerical order and then plotting them based on their percentile position. This makes it easier to see how quickly or slowly the values change from one observation to the next, and whether the increase is smooth or occurs in noticeable jumps. As a result, the chart can help reveal possible structural breaks, clusters, or natural groupings within the data.

As noted previously in **Table 01**, some variables appear to operate within different value ranges, suggesting the possibility of groupings among the sensors. **Visualisation 07** displays the rising order chart for all numeric variables after standardising the values using Z-scores. Standardisation allows the sensors to be compared on a common scale despite differences in their original measurement units.

From this visualisation, most sensor variables follow relatively similar and gradual trends across the percentile range, indicating broadly comparable distributions once standardised. However, *sensor6* displays a noticeable jump shortly after the 50th percentile, where the Z-score increases from approximately -0.5 to +0.5, representing a change of roughly one standard deviation between adjacent ordered values. When plotted by itself in **Visualisation 08**, this jump becomes more obvious.



Visualisation 07: A rising order chart showing all numeric variables after standardisation using Z-scores. Each variable's values are sorted in ascending order and plotted by percentile, allowing distributions to be compared on a common scale. Most numeric variables display relatively smooth increases, while *sensor6* shows a noticeable jump around the median of the distribution. Null values were not enabled in this plot.



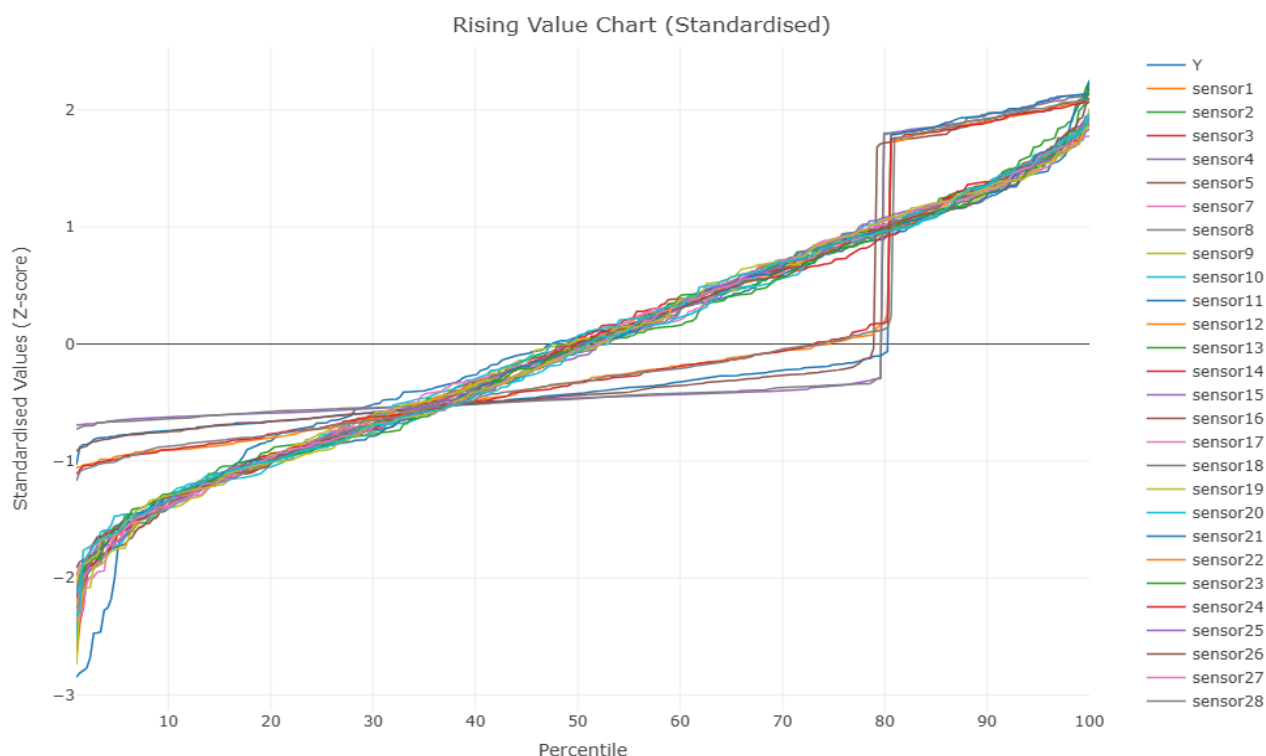
Visualisation 08: A rising order chart displaying the ordered raw values of *sensor6*. The plot highlights a clear jump between approximately 14.70 and 26.13 units, suggesting a potential structural change, grouping within the distribution of this variable, or data quality issue worth exploring.

When *sensor6* is removed from the rising order chart, the standardised distribution appears as shown in **Visualisation 09**. In this plot, a group of seven sensors display a pronounced jump within the distribution. The jump occurs between approximately -0.5 Z-score and just under $+2$ standard deviations, representing a substantial increase across a relatively small percentile range.

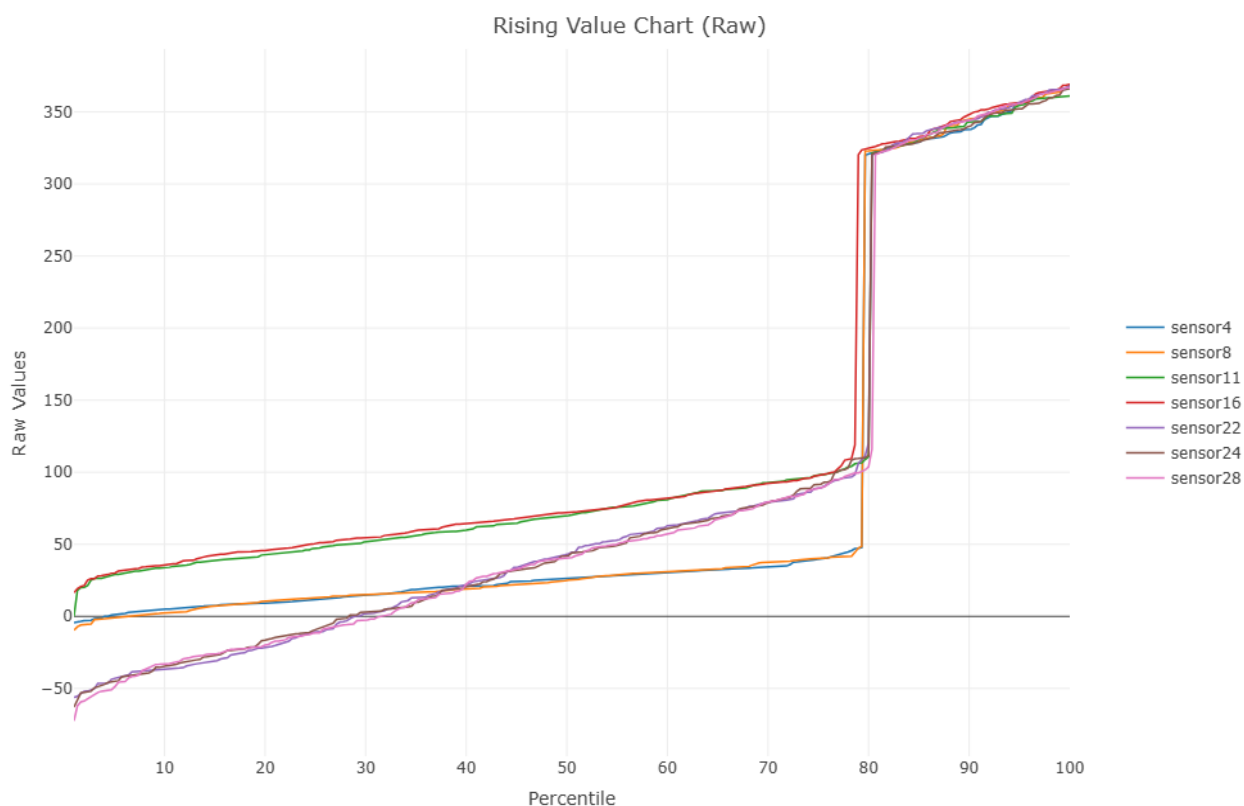
This behaviour was less visible in **Visualisation 07**, most likely due to the influence of *sensor6*, whose large jump and higher proportion of missing values dominated the standardised comparison. Once removed, the patterns of other sensors become more identifiable.

The seven sensors exhibiting this behaviour are *sensor4*, *sensor8*, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28*, which were also previously highlighted in **Visualisation 05**.

To further investigate this pattern, the sensors were plotted individually using their raw measurement values, as shown in **Visualisation 10**. The plot indicates that *sensor4* and *sensor8* begin at relatively low values of around 50 units. In contrast, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28* start at higher baseline levels of approximately 110 units. All sensors show a pronounced jump to values above 320 units. After this jump, the measurements increase more gradually from around the 80th percentile onward, eventually reaching values exceeding 360 units for several sensors.



Visualisation 09: A standardised rising order chart displaying all numeric sensors excluding *sensor6*. The removal of *sensor6* reveals a pronounced jump in the ordered distributions of seven sensors, indicating a potential structural similarity among this group of variables.

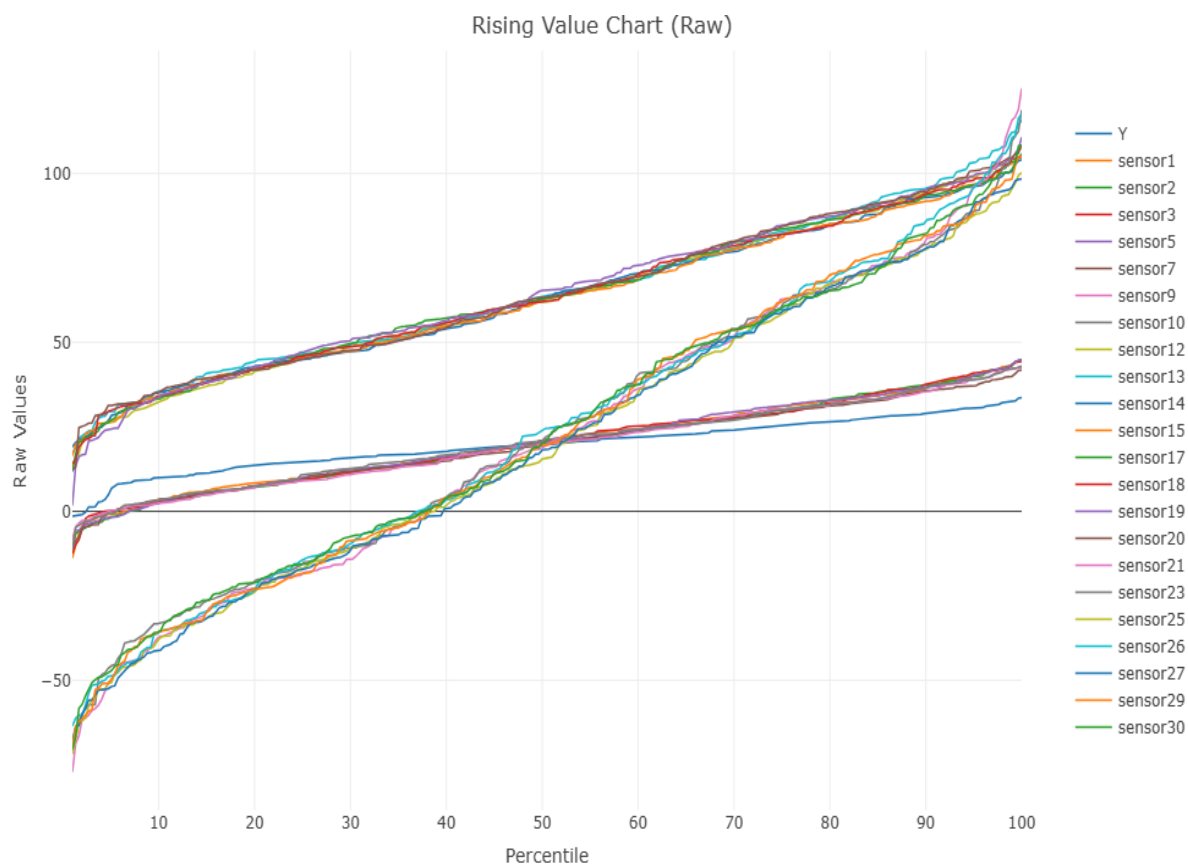


Visualisation 10: Rising order charts showing the raw values for *sensor4*, *sensor8*, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28*. These sensors display a distinct jump in their distributions before gradually increasing toward the upper percentiles, suggesting a shared structural pattern in their measurements.

When *sensor4*, *sensor6*, *sensor7*, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28* are removed, the raw rising order chart reveals three additional distinct groupings among the remaining sensors.

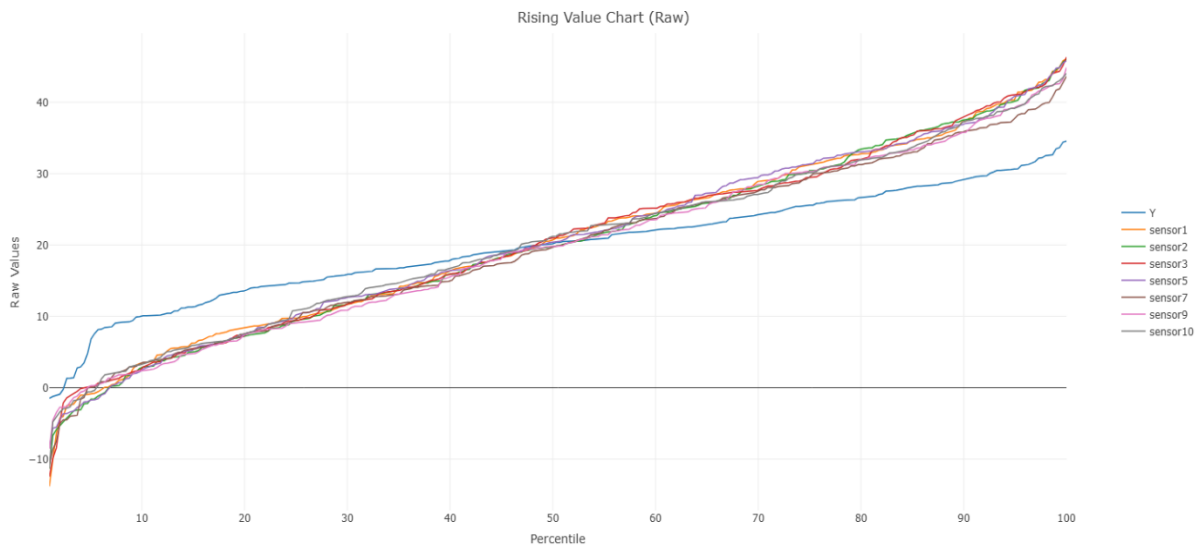
The one group of sensors begins at values close to 0 units and increases gradually across the distribution, reaching a maximum of approximately 50 units. A second group also begins near 0 units but displays a sharp increase within the first 10th percentile before continuing to increase more gradually, eventually reaching maximum values between 100 and 120 units. The third group starts at substantially lower values, around –70 units, before increasing steeply across the distribution and eventually reaching a similar maximum range to the second group.

These distributional patterns suggest the presence of distinct value ranges among subsets of sensors, which can be observed more clearly in **Visualisation 11**, where the sensors are plotted using their raw measurement values.



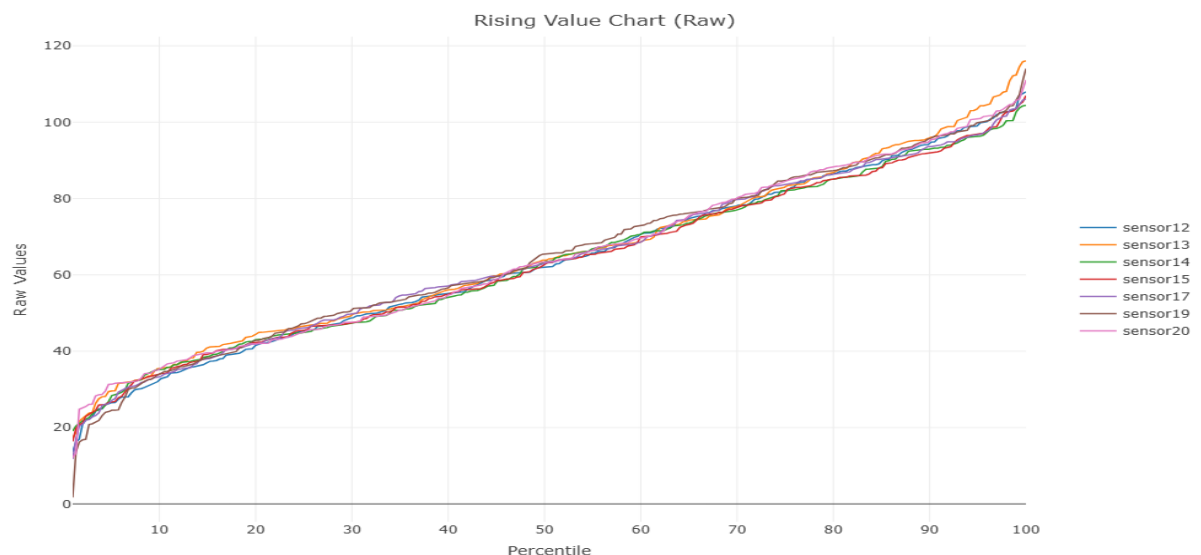
Visualisation 11: A rising order chart displaying the remaining sensor variables after removing *sensor4*, *sensor6*, *sensor7*, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28*. The chart highlights three distinct distributional groupings, with sensors exhibiting different starting ranges and growth patterns across the ordered percentiles.

The first group of variables includes *sensor1*, *sensor2*, *sensor3*, *sensor5*, *sensor7*, *sensor9*, and *sensor10*. Each of these sensors begins at values around –10 units before increasing gradually across the distribution and reaching a maximum of approximately 50 units. Of particular interest is the variable *Y*, which displays a distribution that closely resembles the pattern of these sensors. The variable *Y* begins near 0 units, performs a small step increase before the 10th percentile, and then increases gradually before reaching a maximum of 39.39 units. This pattern can be observed in **Visualisation 12**, which displays the variables using their raw values.



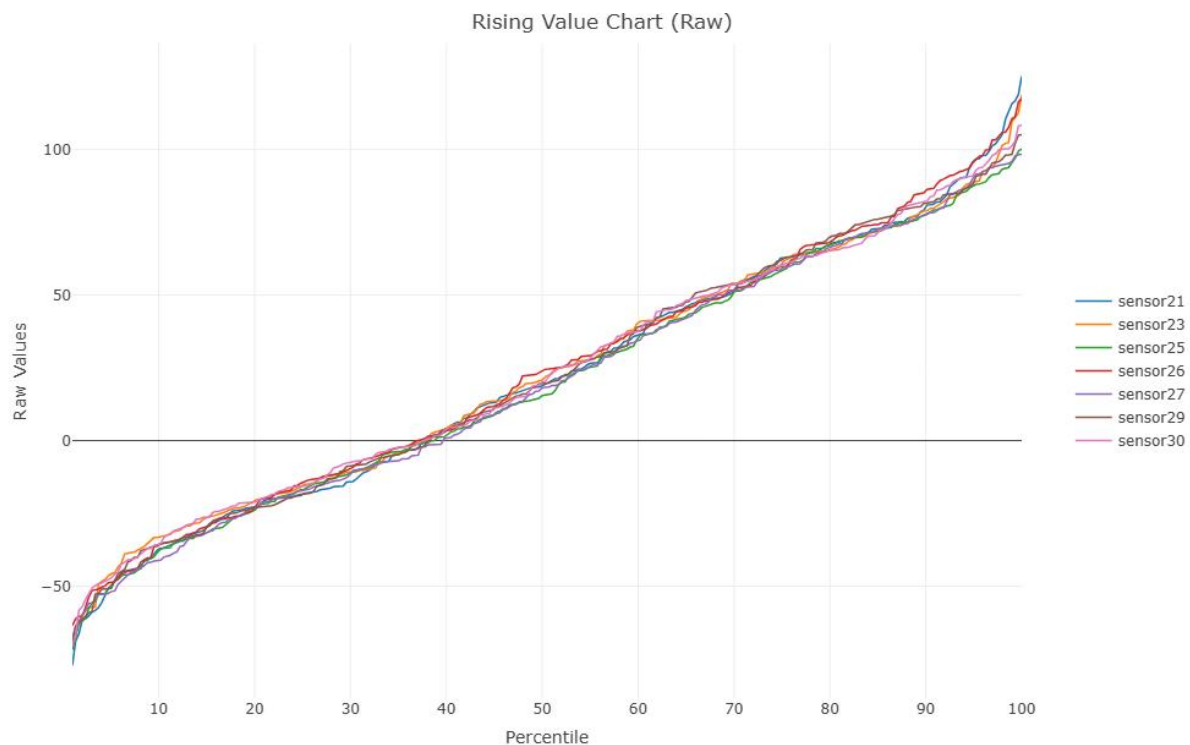
Visualisation 12: A rising order chart displaying the raw values for *sensor1*, *sensor2*, *sensor3*, *sensor5*, *sensor7*, *sensor9*, *sensor10*, and *Y*. These variables begin at values around -10 to 0 units and increase gradually across the ordered percentiles, reaching maximum values of approximately 40–50 units. The variable *Y* follows a similar distributional pattern, with a small increase near the 10th percentile before gradually rising to its maximum value 5-10 units underneath the previously described sensors.

The second group consists of *sensor12*, *sensor13*, *sensor14*, *sensor15*, *sensor17*, *sensor18*, *sensor19*, and *sensor20*. These sensors generally begin between 0 and 20 units, followed by a sharp increase within the first 10th percentile. After this initial rise, the values increase more gradually across the remaining distribution, eventually reaching maximum values between 100 and 120 units. This behaviour is illustrated in **Visualisation 13**, which shows these sensors using their raw values.



Visualisation 13: A rising order chart displaying the raw values for *sensor12*, *sensor13*, *sensor14*, *sensor15*, *sensor17*, *sensor18*, *sensor19*, and *sensor20*. These sensors begin between 0 and 20 units and exhibit a sharp increase within the first 10th percentile, followed by a more gradual increase across the remaining distribution. The sensors eventually reach maximum values between 100 and 120 units, indicating a distinct value range compared with the first group of sensors.

The last group consists of *sensor21*, *sensor23*, *sensor25*, *sensor26*, *sensor27*, *sensor29*, and *sensor30*. These sensors start at substantially lower values, typically around –60 to -70 units, before increasing steeply across the distribution. Despite their lower starting points, these sensors eventually reach maximum values similar to those in the second group, generally within the 100-to-120-unit range. This pattern can be seen in **Visualisation 14**, which presents the sensors using their raw values.



Visualisation 14: A rising order chart displaying the raw values for *sensor21*, *sensor23*, *sensor25*, *sensor26*, *sensor27*, *sensor29*, and *sensor30*. These sensors begin at substantially lower values, typically around –70 to -60 units, before increasing steeply across the distribution. Despite their lower starting points, the sensors eventually reach their maximum values similar to those observed in **Visualisation 13**, generally between 100 to 120 units.

5). Correlation Plot:

Correlation plots are useful for identifying relationships between numerical variables by measuring how strongly they are associated with one another. Most correlation measures range between -1 and $+1$, where -1 represents a perfect negative correlation, $+1$ represents a perfect positive correlation, and values close to 0 indicate little or no linear relationship.

One commonly used method for calculating correlation is Pearson's correlation coefficient. Pearson's correlation measures the strength and direction of a linear relationship between two variables by comparing how their values change relative to their means. It is widely used in exploratory data analysis because it provides a simple and interpretable measure of linear association between variables.

To improve the readability of the correlation plot, the variables can be arranged using angular order of eigenvectors. This ordering method groups variables together based on the similarity of their correlation structure, allowing strongly related variables to appear close to one another in the visualisation. As a result, clusters of highly correlated variables become easier to identify.

Visualisation 15 was created to examine how the numerical variables correlate with one another. The plot reveals several distinct groups of highly correlated sensors, which closely align with the groupings previously identified in the rising order chart analysis (**Visualisations 10, 12, 13, and 14**).

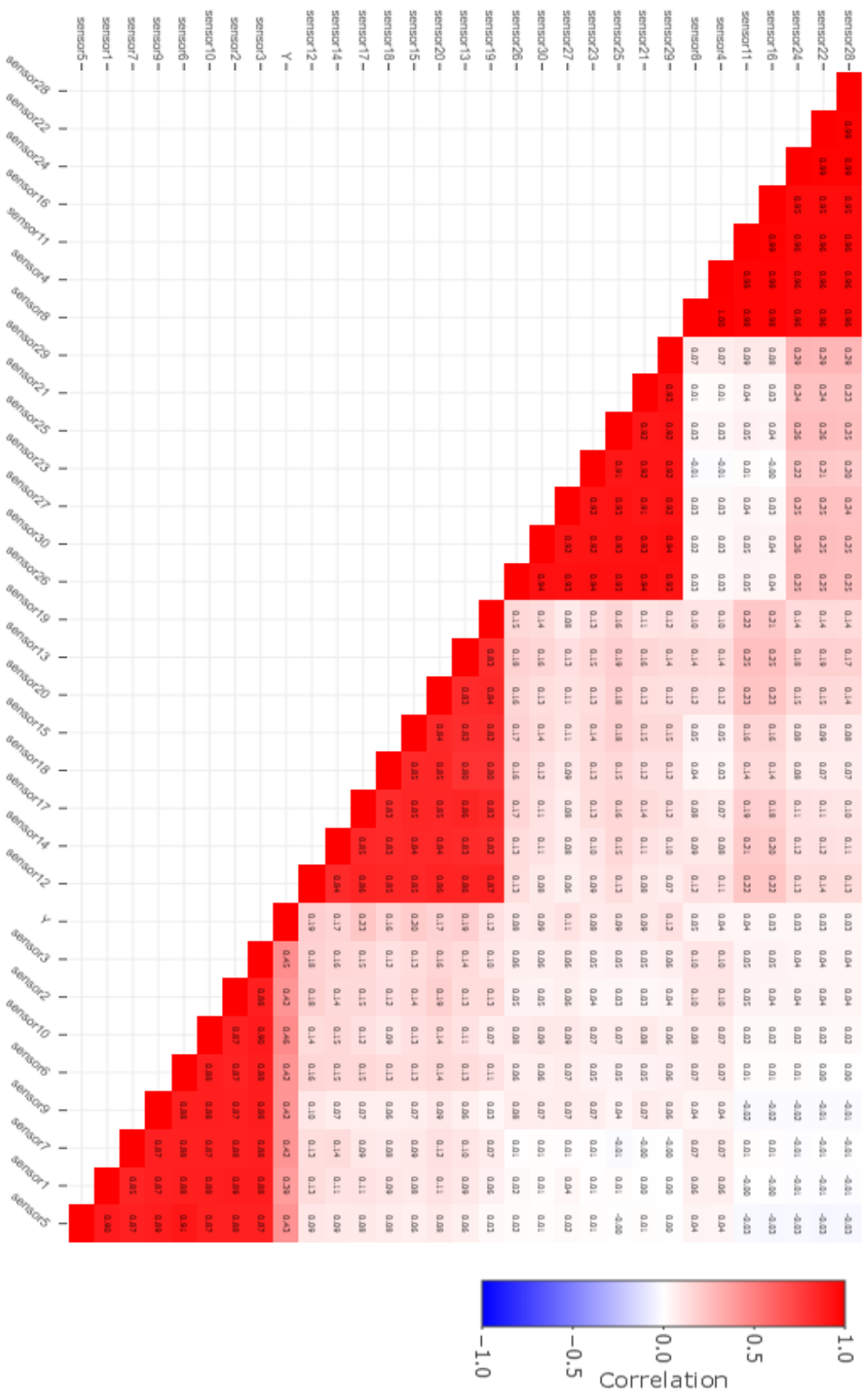
Four main correlation groups can be observed:

- Group A: *sensor4, sensor8, sensor11, sensor16, sensor22, sensor24, and sensor28*
- Group B: *sensor21, sensor23, sensor25, sensor26, sensor27, sensor29, and sensor30*
- Group C: *sensor12, sensor13, sensor14, sensor15, sensor17, sensor18, sensor19, and sensor20*
- Group D: *sensor1, sensor2, sensor3, sensor5, sensor6, sensor7, sensor9, and sensor10*

Within the first two groups, many of the sensor pairs exhibit very strong positive correlations exceeding $r = 0.9$, indicating that these sensors behave very similarly across observations. This supports the structural groupings previously identified through the rising order charts. Amongst the last two groups, there was also strong positive correlation but only exceeding $r = 0.8$ correlation.

The only variable that does not strongly align with any of these sensor clusters is the response variable Y . When examined using Pearson's correlation coefficient, Y shows its strongest relationships with Group D sensors. However, these correlations are moderate at best rather than strong, ranging between $r = 0.387$ and $r = 0.456$.

Correlation Heat Map
Method: Pearson | Order: Angular Order of Eigenvectors



Visualisation 15: A correlation matrix displaying Pearson correlation coefficients between all numeric variables. The variables are ordered using angular ordering of eigenvectors, allowing strongly correlated variables to appear closer together in the plot. Several clusters of sensors with high positive correlations (above $r = 0.8$ and $r = 0.9$) can be observed, which correspond to the sensor groupings previously identified through the rising order chart analysis. Null values were not enabled in this plot.

While Pearson's correlation provides a useful measure of linear association, it can be sensitive to outliers and extreme values. Given that such values were identified in earlier visualisations (particularly among Group A sensors), an alternative correlation approach was also considered.

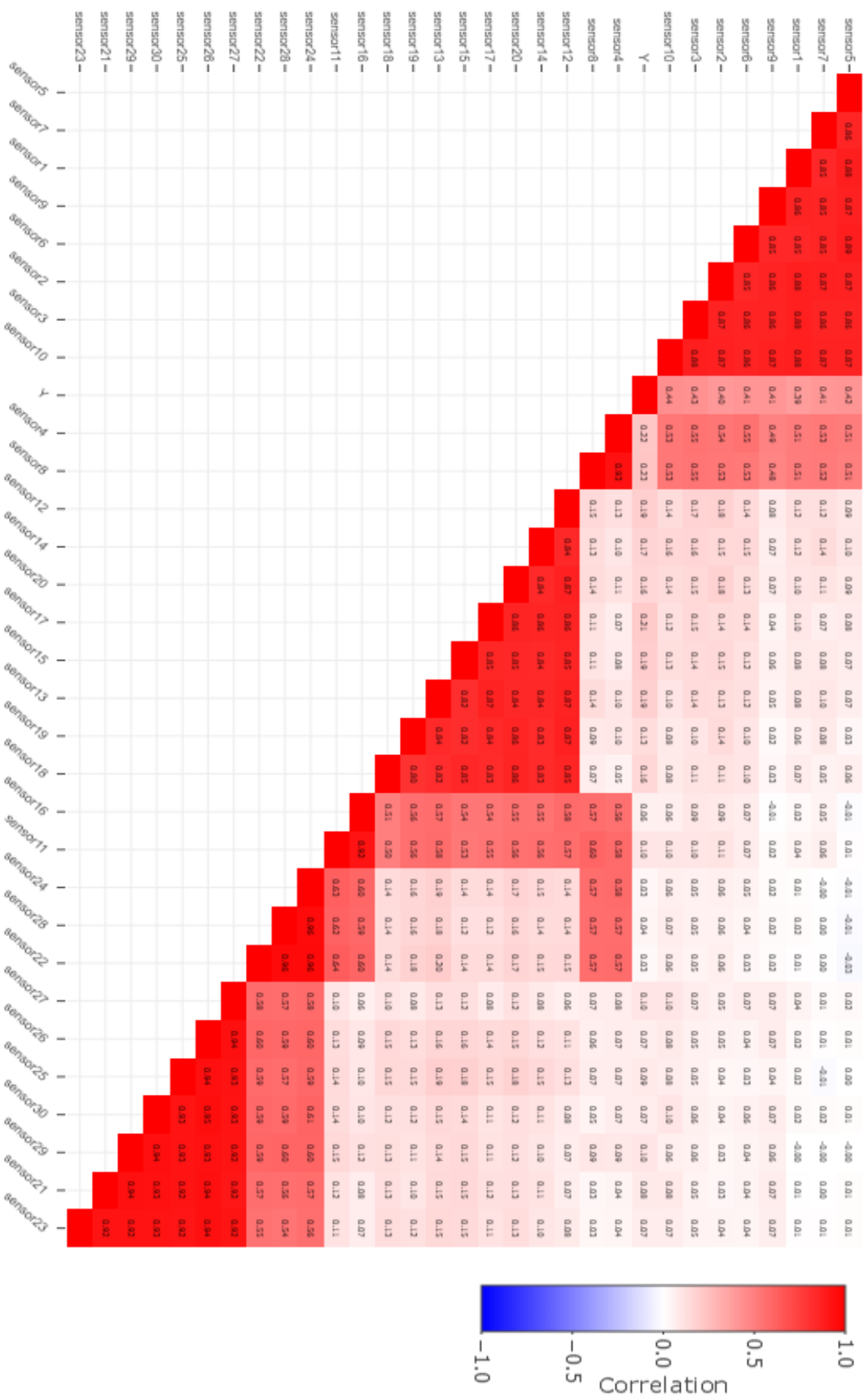
Spearman's rank correlation is another commonly used method. Unlike Pearson's correlation, which assesses the strength of a linear relationship between variables, Spearman's correlation evaluates the strength of a monotonic relationship based on the ranked values of the data. Because it relies on ranks rather than raw measurements, it is generally more robust to outliers and extreme observations, as unusually large or small values have less influence on the overall correlation estimate.

To further explore relationships between variables, all 31 numeric variables were analysed using a Spearman correlation matrix. As with the Pearson analysis, the variables were ordered using the angular order of eigenvectors to maintain consistency in the visual comparison and to help reveal any clustering patterns. The Spearman results identified three main sensor groupings, which broadly align with the structural patterns observed earlier in the rising order charts.

- Group E consists of *sensor1*, *sensor2*, *sensor3*, *sensor4*, *sensor5*, *sensor6*, *sensor7*, *sensor8*, *sensor9*, *sensor10*, and *Y*. Within this group, *sensor1*, *sensor2*, *sensor3*, *sensor5*, *sensor6*, *sensor7*, and *sensor9* show strong positive correlations with one another, typically ranging from approximately $r = 0.89$ to $r = 0.94$. *sensor4* and *sensor8* display weaker associations with the rest of the group, with correlations generally between $r = 0.48$ and $r = 0.55$. The response variable *Y* shows moderate correlations with sensors in this group, with values ranging from about $r = 0.39$ to $r = 0.44$.
- Group F includes *sensor11*, *sensor12*, *sensor13*, *sensor14*, *sensor15*, *sensor16*, *sensor17*, *sensor18*, *sensor19*, and *sensor20*. Within this group, *sensor12*, *sensor13*, *sensor14*, *sensor15*, *sensor17*, *sensor18*, and *sensor19* show strong correlations with one another, typically ranging between $r = 0.80$ and $r = 0.87$. *sensor11* and *sensor16* show moderate correlations with the rest of the group, with correlation values generally falling between $r = 0.50$ and $r = 0.58$.
- Group G contains of *sensor21*, *sensor22*, *sensor23*, *sensor24*, *sensor25*, *sensor26*, *sensor27*, *sensor28*, *sensor29*, and *sensor30*. Within this group, *sensor21*, *sensor23*, *sensor25*, *sensor26*, *sensor27*, *sensor29*, and *sensor30* exhibit strong correlations with one another, typically between $r = 0.91$ and $r = 0.94$. *sensor22*, *sensor24*, and *sensor28* show moderately strong correlations with the rest of the group, with correlations commonly around $r = 0.60$.

These findings are summarised in **Visualisation 16**, which presents the Spearman correlation matrix and highlights the clustering of sensors into three distinct correlation groups. The reorganisation of sensors into Groups E, F, and G under Spearman correlation reflects the method's lower sensitivity to extreme values. As a result, some sensors that previously appeared structurally distinct (such as those seen in Group A) align more closely with the broader correlation structure when ranked relationships are considered.

Correlation Heat Map
Method: Spearman | Order: Angular Order of Eigenvectors



Visualisation 16: A correlation matrix displaying Spearman rank correlation coefficients between all numeric variables. The variables are ordered using angular ordering of eigenvectors, allowing clusters of strongly related sensors to appear adjacent in the visualisation. Three main sensor groupings can be observed, with several sensors exhibiting strong positive correlations above $r = 0.9$, indicating similar measurement behaviour within these groups. Null values were not enabled in this plot.

6). Time Series Plot (Part Two):

Based on the findings from the rising order charts and correlation analyses, it would be useful to examine how the sensors behaved over time using the identified sensor groupings. Time series plots were created to observe whether sensors within each group exhibited similar temporal patterns.

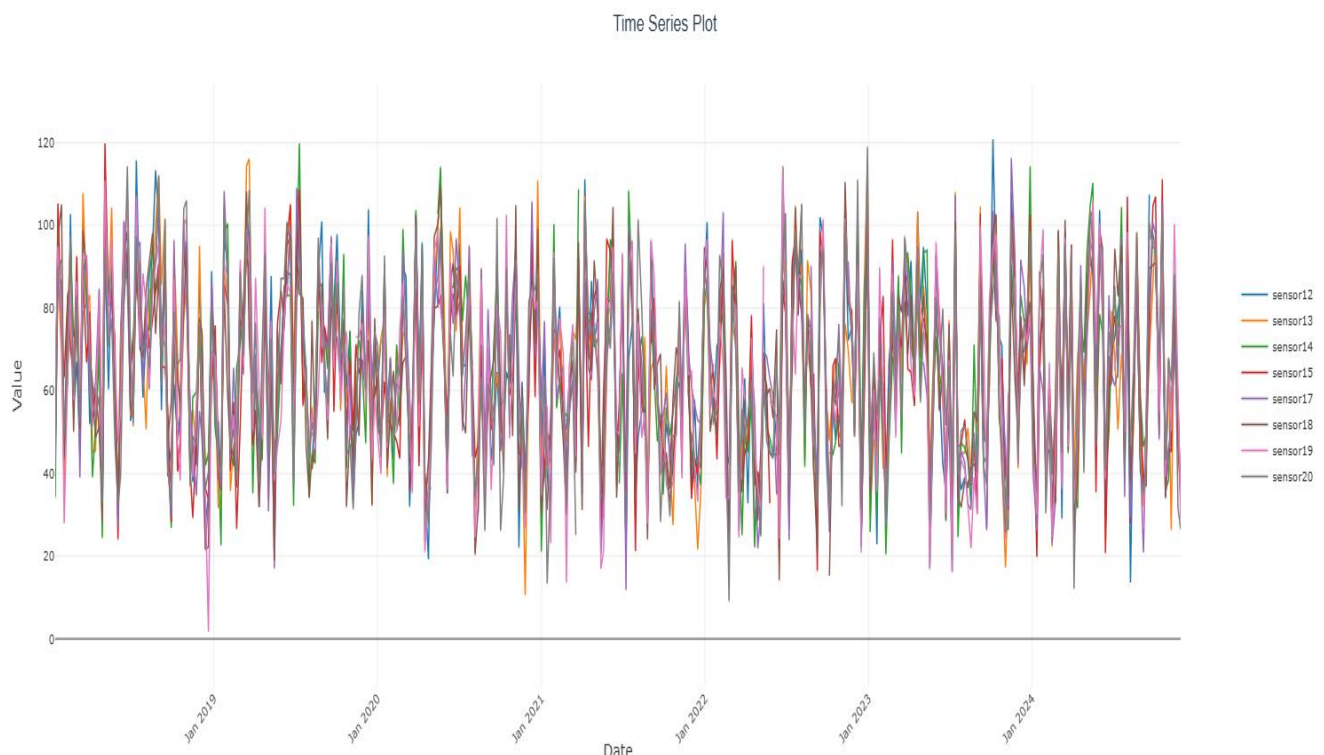
The following groups were analysed:

- Group B: *sensor21, sensor23, sensor25, sensor26, sensor27, sensor29, and sensor30*
- Group C: *sensor12, sensor13, sensor14, sensor15, sensor17, sensor18, sensor19, and sensor20*
- Group D: *sensor1, sensor2, sensor3, sensor5, sensor6, sensor7, sensor9, and sensor10*

Group A had already been examined earlier (**Visualisation 5**) so the focus of this section is on the remaining three above groups.

An important observation from these plots is that sensors within each group display very similar temporal patterns and magnitudes over time. Within each group, the sensor readings tend to rise and fall together at similar levels, indicating a high level of synchronisation in their behaviour. This observation is consistent with the strong correlations previously identified in the Pearson and Spearman correlation analyses.

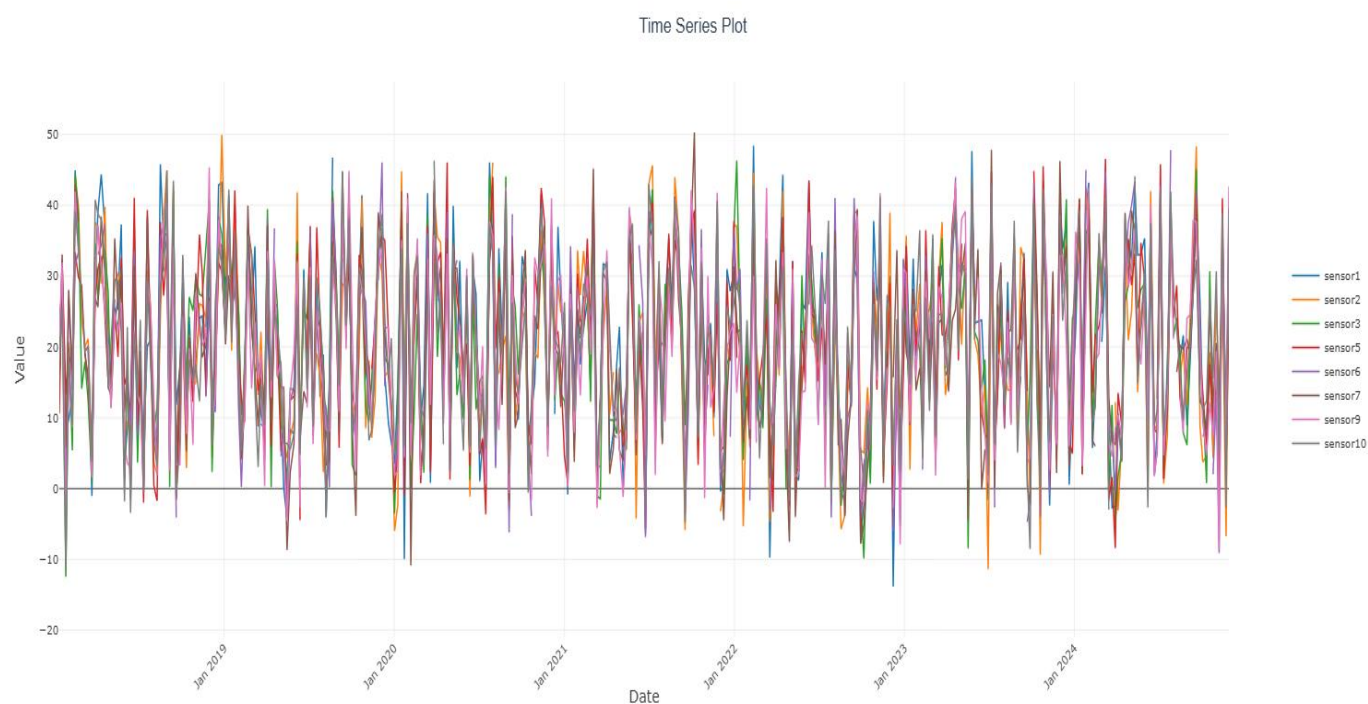
The time series plots illustrating these patterns with their raw values are shown in **Visualisation 17** (Group B), **Visualisation 18** (Group C), and **Visualisation 19** (Group D).



Visualisation 17: A time series plot showing the readings for *sensor12, sensor13, sensor14, sensor15, sensor17, sensor18, sensor19, and sensor20*. These sensors follow closely aligned trends throughout the observation period, reinforcing the strong correlations observed between them.



Visualisation 18: A time series plot showing the sensor readings for *sensor21*, *sensor23*, *sensor25*, *sensor26*, *sensor27*, *sensor29*, and *sensor30* across the observation period. Sensors within this group display very similar temporal patterns, with values increasing and decreasing together over time.



Visualisation 19: A time series plot displaying *sensor1*, *sensor2*, *sensor3*, *sensor5*, *sensor6*, *sensor7*, *sensor9*, and *sensor10* over time. Sensors within this group display very similar temporal patterns, with values increasing and decreasing together over time.

7). Boxplot (Part One):

Given the success in using the correlation plot relationships, the next set of visualisations examines the distribution of the numeric variables using boxplots. Boxplots are useful for summarising the key distributional characteristics of a variable, including the minimum value, first quartile (Q1), median, third quartile (Q3), and maximum value. They also display the interquartile range (IQR), which represents the spread of the middle 50% of the data. In addition, boxplots are particularly helpful for identifying potential outliers, which are typically defined as observations that fall beyond 1.5 times the interquartile range ($1.5 \times \text{IQR}$) from the quartiles.

The boxplots in **Visualisation 20** were generated using the raw sensor values, without any scaling or centring applied. When viewed in this form, the boxplots reveal three distinct value ranges across the sensors, which closely follow the groupings previously identified in the Spearman correlation analysis (Groups E, F, and G).

Another important feature observed in the boxplots is the presence of a large number of outliers among a subset of sensors. Using the standard $1.5 \times \text{IQR}$ threshold, *sensor4*, *sensor8*, *sensor11*, *sensor16*, *sensor22*, *sensor24*, and *sensor28* collectively produce 440 identified outliers. These sensors correspond to the group previously identified as having large jumps in the rising order charts, further reinforcing the structural differences observed among the sensor groups.

In contrast, the only other variable that displayed an outlier was the response variable *Y*, with a single observation identified at -5.71 units using the $1.5 \times \text{IQR}$ rule. This indicates that, aside from the Group A sensors, both *Y* and the remaining sensor variables are much less affected by extreme values. Overall, their distributions appear relatively stable when compared with the higher prevalence of outliers observed among the Group A sensors.

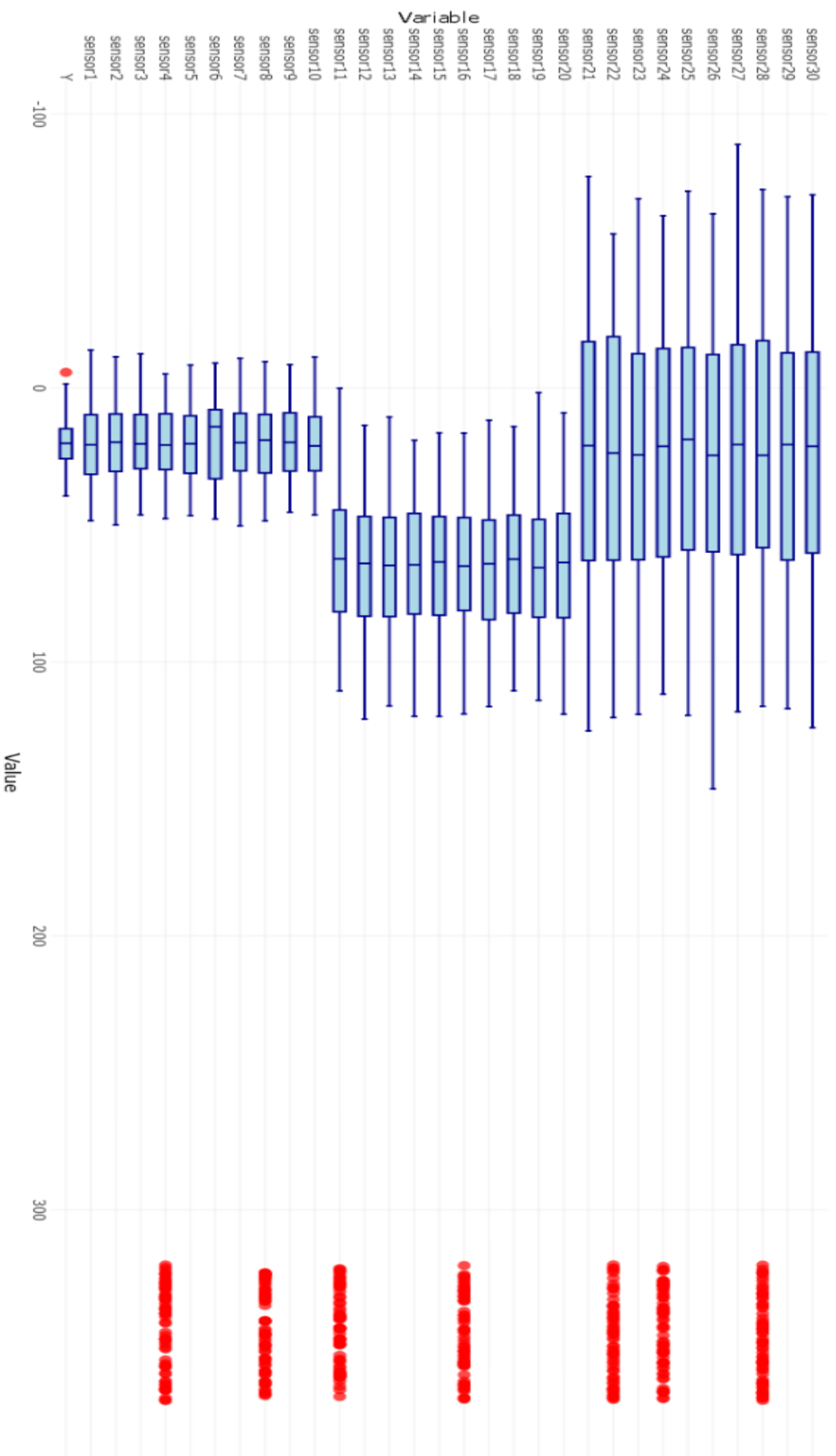
One advantage of boxplots is that the IQR multiplier can be adjusted to explore how sensitive the outlier classification is to the chosen threshold. When increasing the multiplier beyond the standard $1.5 \times \text{IQR}$, the number of detected outliers gradually decreases. Outliers only begin to substantially reduce when the multiplier reaches approximately $2.2 \times \text{IQR}$.

At approximately $2.8 \times \text{IQR}$, the sensors *sensor22*, *sensor24*, and *sensor28* no longer contain any identified outliers. When the multiplier increases further to around $5.9 \times \text{IQR}$, sensors *sensor11* and *sensor16* also absorb their previously identified outliers. Finally, at approximately $12.1 \times \text{IQR}$, sensors *sensor4* and *sensor8* absorb all remaining outliers. All these findings can be demonstrated with **Visualisations 21, 22, and 23**.

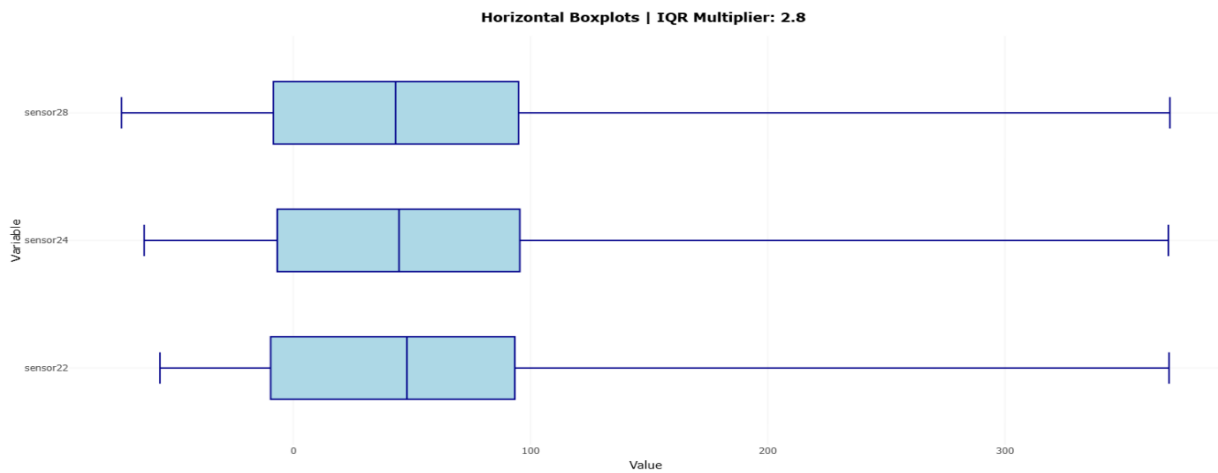
This behaviour suggests that the observations initially identified as outliers are not necessarily isolated anomalies but may instead represent values belonging to a different measurement range within this group of sensors. This finding reinforces the structural separation between the sensor groups, consistent with the groupings previously identified through the rising order charts and correlation analyses.

Beyond the outlier analysis, most variables within Groups E, F, and G follow similar statistical patterns to one another. However, two exceptions are observed within **Visualisation 20**. Those exceptions are *sensor6* does not follow the rest of Group E, with a much smaller median value that is 0.5 standard deviation units lower from the other variables in the group, while maintaining the same IQR and range. Additionally, variable *Y* displays a smaller IQR and range compared to Group E but has a similar median value. These findings are further collaborated with **Table 01**.

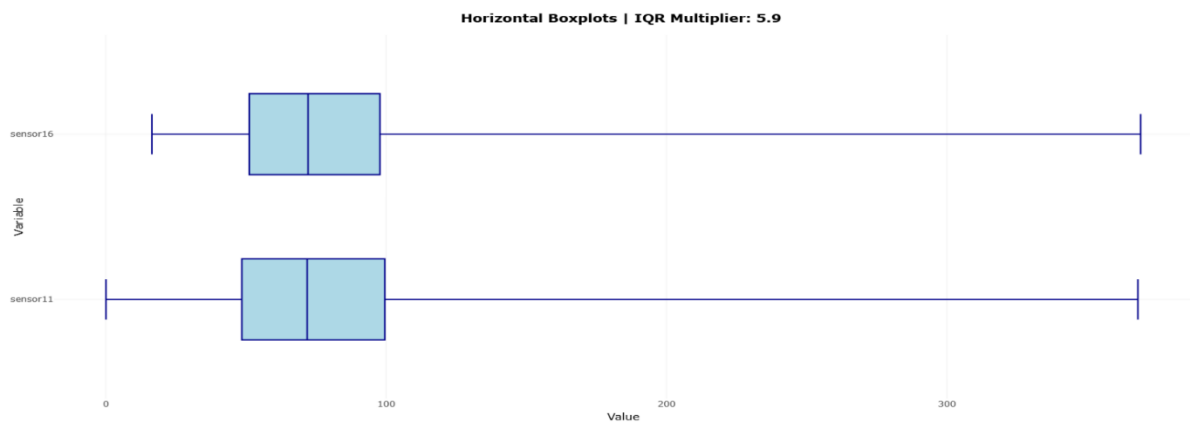
Horizontal Boxplots | IQR Multiplier: 1.5



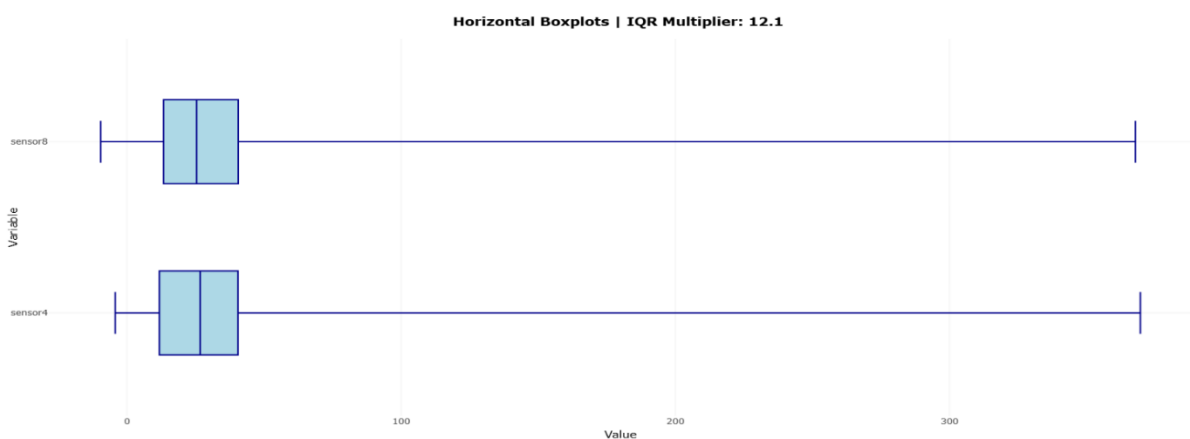
Visualisation 20: A boxplot visualisation displaying the distribution of all numeric sensor variables and the response variable Y using raw values. The plots summarise the minimum, first quartile, median, third quartile, and maximum values, while also highlighting potential outliers using the standard $1.5 \times \text{IQR}$ threshold. The visualisation reveals three distinct sensor value ranges, consistent with the previously identified correlation groupings of Groups E, F and G. A substantial number of potential outliers (441 outliers) are present only within sensors 4, 8, 11, 16, 22, 24, and 28. In contrast, the response variable Y displays only one potential outlier (a value of -5.71 units), indicating a relatively stable distribution. This plot was constructed with NULL values enabled. The only substantive difference with NULL values not being enabled in the inclusion of the Y value outlier.



Visualisation 21: Displays the point at which the IQR multiplier reaches approximately 2.8 times the standard threshold. At this level, sensors *sensor22*, *sensor24*, and *sensor28* no longer contain any identified outliers, indicating that their most extreme values have been absorbed back into the main distribution.



Visualisation 22: Displays the point at which the IQR multiplier increases further to approximately 5.9 times the standard threshold. At this level, sensors *sensor11* and *sensor16* also absorb their previously identified outliers, leaving only a subset of sensors with extreme values remaining.



Visualisation 23: Displays the point at which the IQR multiplier reaches approximately 12.1 times the standard threshold. At this final level, sensors *sensor4* and *sensor8* absorb all remaining outliers, meaning no sensors now contain values beyond the specified IQR threshold.

8). ggPairs Plot:

Given the relationships identified in the previous visualisations, it is useful to further examine how these variables interact with one another. One way to explore these interactions is using a ggPairs plot, which allows multiple variables to be compared simultaneously.

A ggPairs plot presents pairwise comparisons between variables within a dataset. Scatterplots in the lower panels display relationships between pairs of numeric variables, while the upper panel's report the corresponding correlation coefficients. Density plots along the diagonal illustrate the distribution of each variable. This structure allows relationships, clustering patterns, and distributional characteristics to be examined in a single visualisation.

The ggPairs plot in **Visualisation 24** compares the variables *Y*, *sensor1*, *sensor4*, *sensor6*, *sensor11*, *sensor12*, *sensor21*, and *sensor22*. These variables were selected as representative examples of the previously identified sensor groupings (Groups A–G), and to illustrate how the response variable *Y* and distinct *sensor6* relates to several of the sensors.

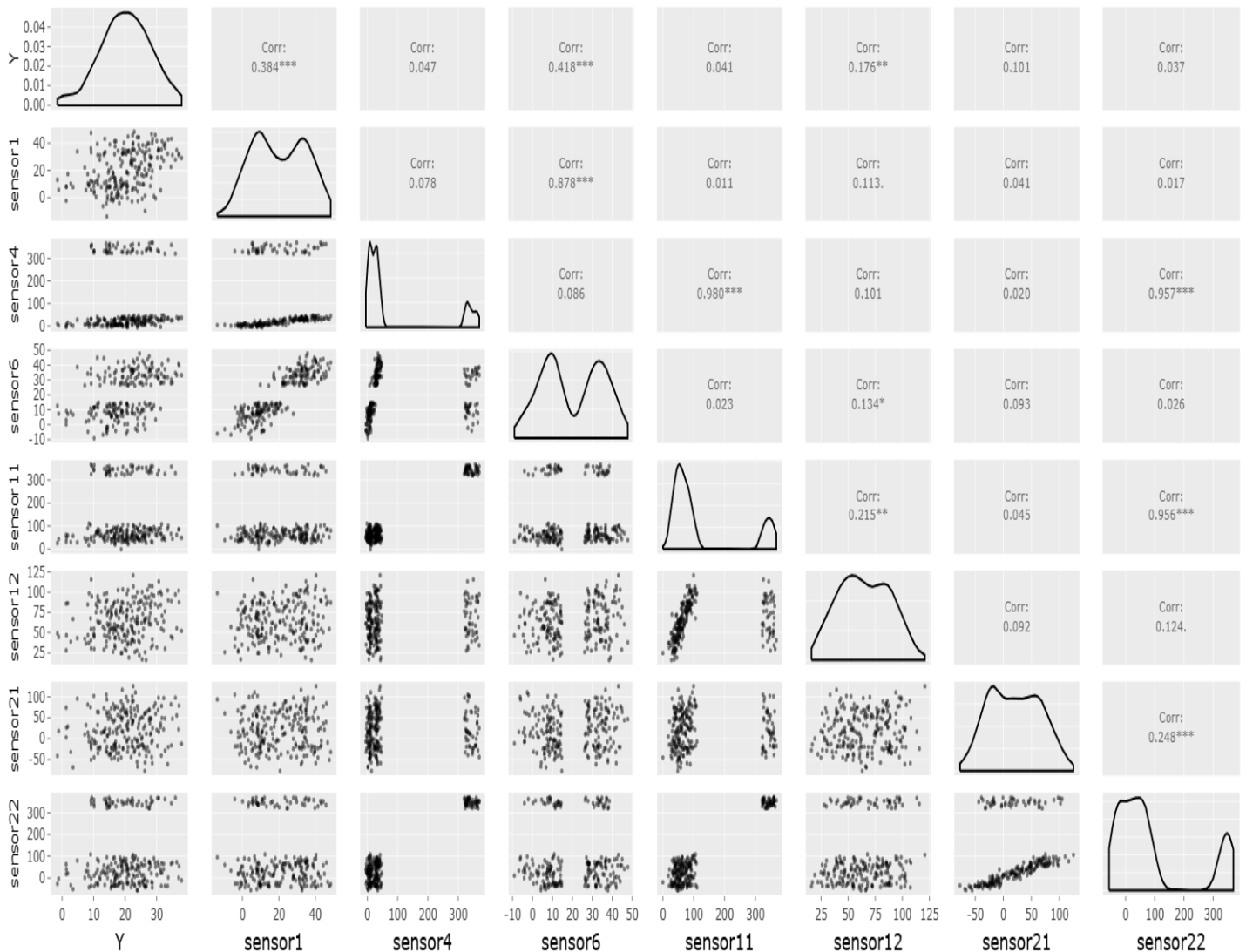
Several strong relationships between sensors are clearly visible. In particular, *sensor4*, *sensor11*, and *sensor22* exhibit extremely strong positive correlations ($r = 0.98$ between *sensor4* and *sensor11*, $r = 0.96$ between *sensor4* and *sensor22*, and $r = 0.96$ between *sensor11* and *sensor22*). These relationships were previously identified in the correlation analysis and correspond to the sensors classified within Group A. The scatterplots show two clearly separated clusters of observations, suggesting that these sensors operate within two distinct value regimes rather than forming a single continuous distribution.

Moderate positive relationships are also observed between the response variable *Y* and several sensors. The strongest associations occur with *sensor6* ($r = 0.42$) and *sensor1* ($r = 0.38$). The corresponding scatterplots display a general upward trend, although the points remain relatively dispersed, indicating that while these variables are related to *Y*, the relationships are not strongly deterministic.

Distinct grouping patterns are also visible in relationships involving *sensor6*. The scatterplots show two noticeable bands of observations, which aligns with the distribution jump previously observed in the rising order charts. This suggests that *sensor6* may contain two underlying value ranges or is missing too much data to contribute to this gap. Given *sensor6*'s high missingness (28.9%), this gap is more likely attributable to missing data than to a genuine bimodal distribution. However, an appropriate investigation is warranted to get a confirmation.

The density plots along the diagonal provide further insight into the distributions of the variables. Several sensors, particularly *sensor4*, *sensor11*, and *sensor22*, display bimodal density shapes. These distributions are consistent with the clustered scatterplot patterns and further support the presence of two distinct ranges of values within the data. In contrast, variables such as *sensor12* and *sensor21* appear more smoothly distributed and show weaker relationships with the other sensors, indicating that they may behave more independently.

GGpairs Plot



Visualisation 24: A ggPairs plot showing pairwise relationships between *Y*, *sensor1*, *sensor4*, *sensor6*, *sensor11*, *sensor12*, *sensor21*, and *sensor22*. Scatterplots in the lower panels illustrate relationships and clustering patterns between variables, while the upper panels display the corresponding correlation coefficients. Density plots along the diagonal show the distribution of each variable. Strong correlations and clustered structures are visible among *sensor4*, *sensor11*, and *sensor22*, while *Y* shows moderate relationships with *sensor1* and *sensor6*.

To further investigate the clustering patterns observed in **Visualisation 24**, the ggPairs plot was reproduced with observations coloured according to the categorical variable *Operator*, as shown in **Visualisation 25**. This allows the relationships between variables to be examined across different operator groups.

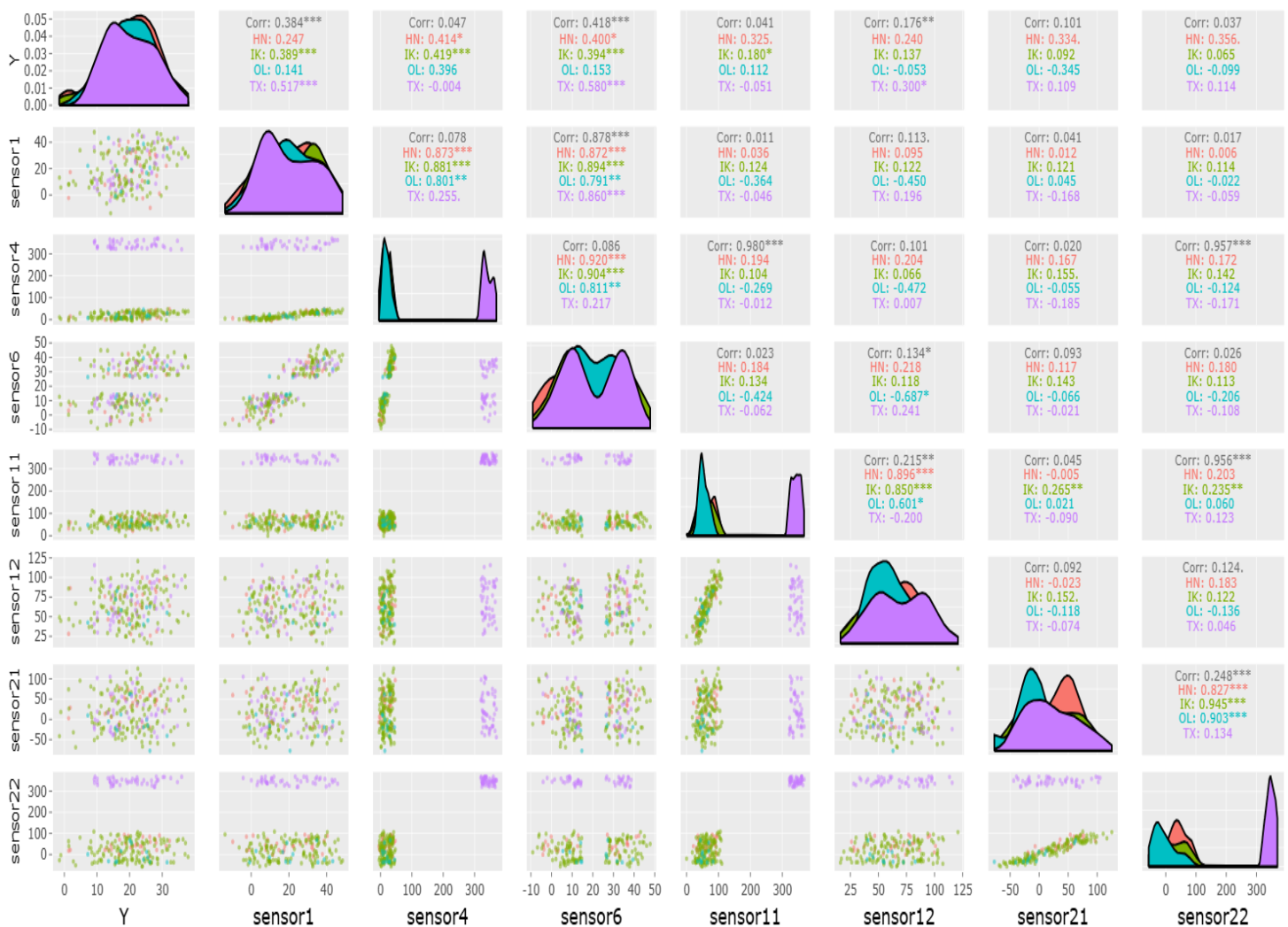
When coloured by *operator*, it becomes clear that the *TX* operator (shown in purple) plays a significant role in the clustering patterns observed for several sensors. In particular, the higher-value cluster present in the relationships between *sensor4*, *sensor11*, and *sensor22* is all composed of *TX* observations. These purple points form a distinct grouping at values above approximately 300 units,

while observations from other operators remain concentrated within the lower value ranges. This suggests that the extreme values contributing to the bimodal distributions of the Group A sensors are primarily associated with measurements recorded under the TX operator. This was checked with **Visualisation 26** that displayed all other Group A variables.

All other sensors show substantially less separation between operators. Their scatterplots display considerable overlap across operator groups, and their distributions remain relatively unimodal. This suggests that these variables are less influenced by operator-specific effects.

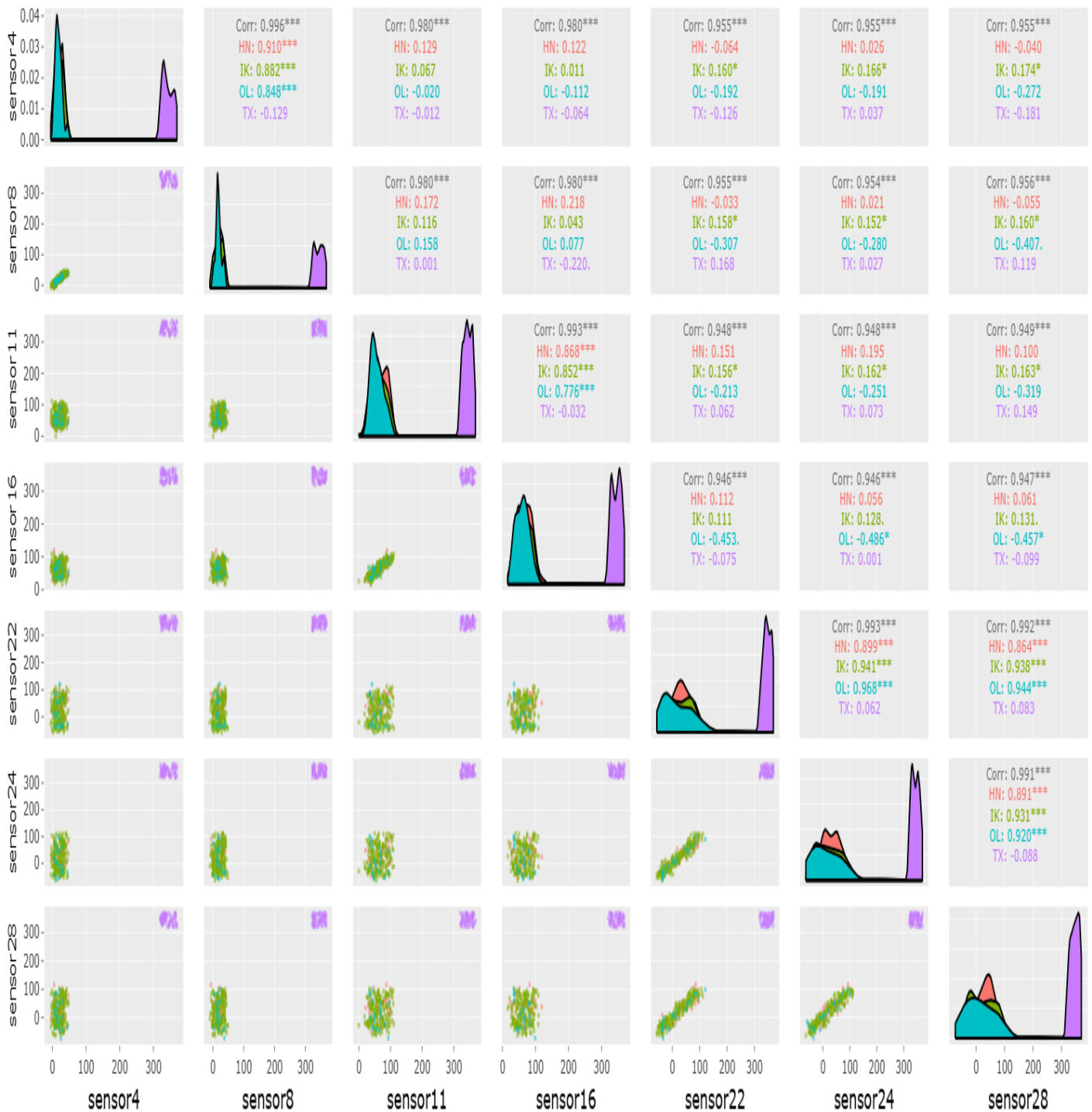
Additional categorical variables were also explored as potential grouping factors. However, none produced a level of separation in the numerical sensor values comparable to that observed when colouring by *operator*, particularly for the Group A sensors. This indicates that *operator* is a key factor associated with several of the distinct clustering patterns observed in the dataset.

GGpairs Plot - Colored by Operator



Visualisation 25: A ggPairs plot of the same variables coloured by *operator*. The plot reveals that the TX operator (purple) forms a distinct cluster at higher values for *sensor4*, *sensor11*, and *sensor22*, contributing to the bimodal distributions observed for these Group A sensors. Other operators remain concentrated in lower value ranges. While several other categorical variables were examined, none produced a separation of the numerical variables as great of an effect or clearly as *Operator* variable.

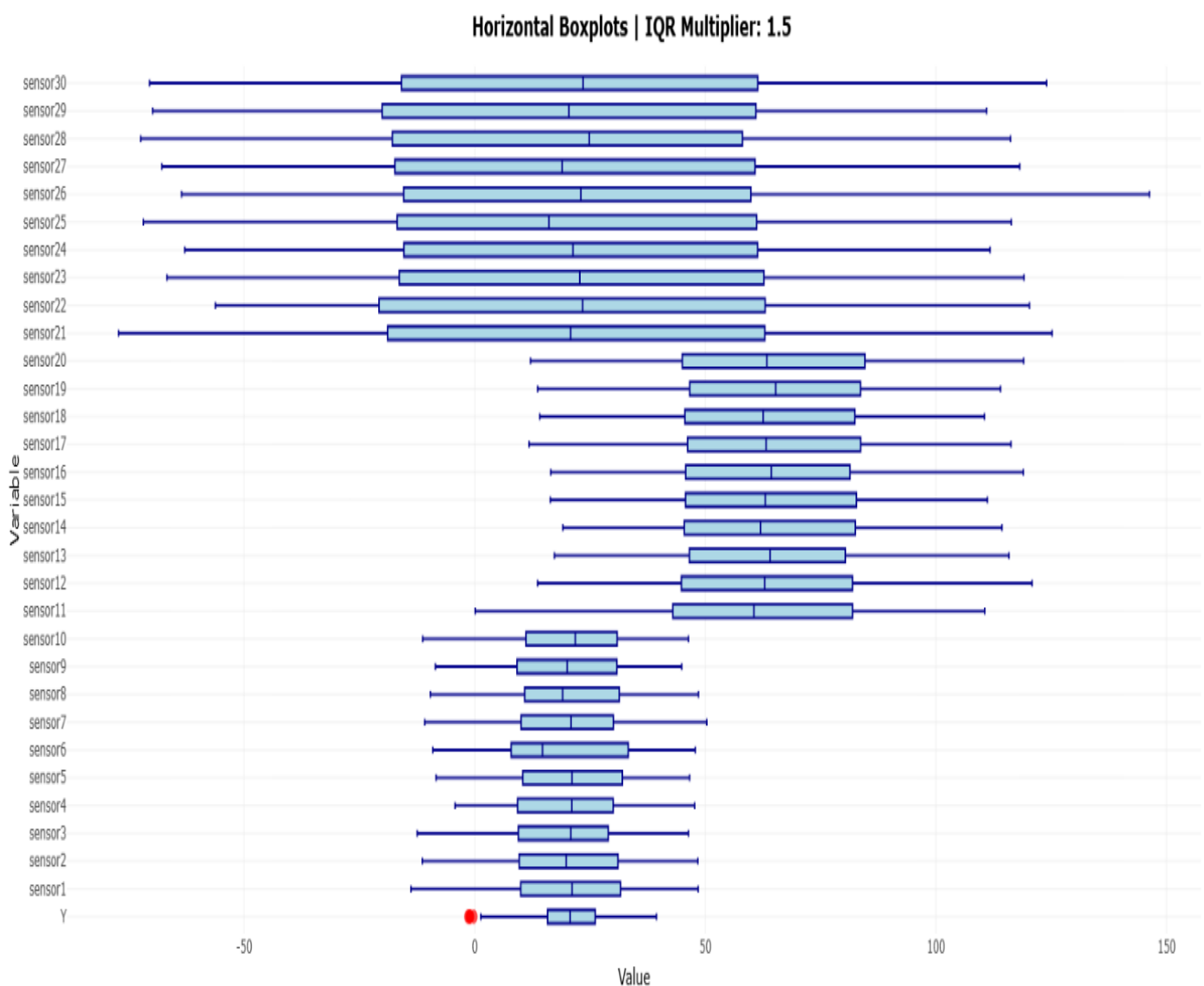
GGpairs Plot - Colored by Operator



Visualisation 26: A ggPairs plot all the Group A (*sensor4*, *sensor8*, *sensor11*, *sensor16* and *sensor22*, *sensor24*, and *sensor28*) variables coloured by *Operator* reveals a distinct clustering pattern. The *TX* operator (purple) consistently appears as a discrete cluster in the extreme top-right quadrant across all pairwise combinations. This pattern was verified as unique to Group A variables, as no other numerical variables produced this top-right quadrant separation. The visualisation confirms that *TX* operator exhibits systematically elevated readings across all Group A sensors.

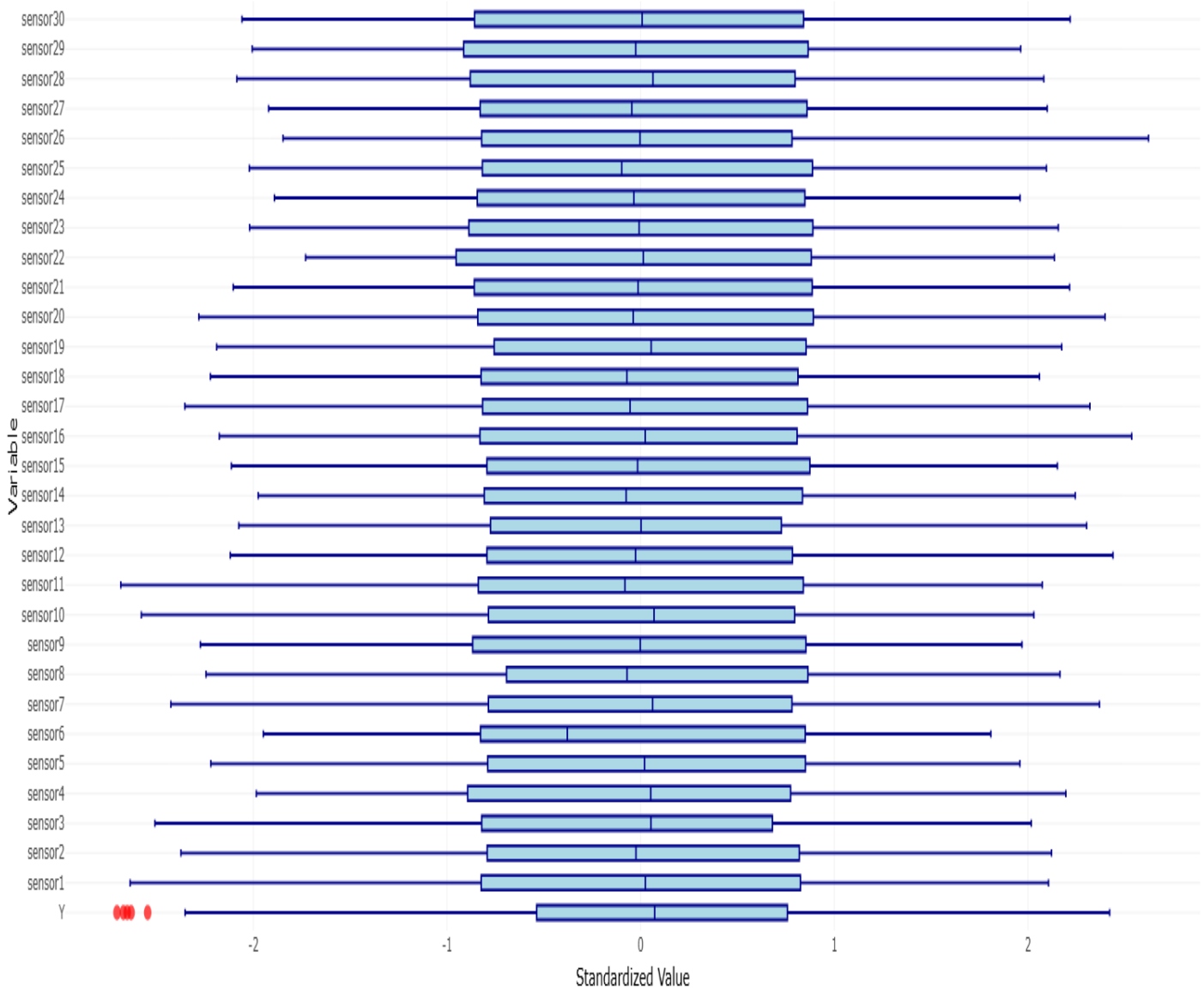
9). Boxplot (Part Two):

Returning to the boxplot analysis, when the TX operator values are removed from the dataset, the remaining numerical variables form more uniform groupings in line with Groups E, F, and G. With these outliers excluded, the data behaves similarly to the correlation structure observed in **Visualisation 16**. **Visualisation 27** demonstrates this by presenting boxplots of all numerical variables with TX operator excluded, revealing three distinct groups without the extreme values at the upper range. **Visualisation 28** presents the same boxplots but centred and scaled, highlighting the overall homogeneity between variables. Notable exceptions include *sensor6*, which displays a smaller median value at approximately -0.5 standard deviations from the others, and the *Y* variable, which contains five outliers ranging from -2.7 to -2.55 units.



Visualisation 27: Boxplots displaying all numerical variables after removal of the TX operator values demonstrate more uniform groupings across the dataset. The exclusion of these outlier values reveals three distinct distributions as described as Groups E - G with no extreme values present at the upper range. This visualisation confirms that the TX operator was responsible for the previously observed skew, and the remaining data now exhibits the homogeneous behaviour also identified in the correlation plot of Visualisation 16.

Horizontal Boxplots | IQR Multiplier: 1.5 (Centered & Scaled)



Visualisation 28: Centred and scaled boxplots of all numerical variables (*TX operator* removed) demonstrate the relative homogeneity between variables. All distributions display comparable ranges and interquartile intervals when standardised. However, there are two exceptions that are evident. One is *sensor6* having a smaller median value at approximately -0.5 standard deviations below the other variables, and the *Y* variable contains five detected outliers ranging from -2.7 to -2.55 units. Outside of these minor variations aside, the visualisation confirms consistent behaviour across the remaining sensor groups.

10). Time Series Plot (Part Three):

Having established the *TX operator*'s distinctive influence on Group A sensors, a time series analysis was conducted to examine its temporal contribution. **Visualisation 29** presents a time series plot with *TX operator* values removed, revealing a noticeable gap during the period previously containing the extreme high values. Alternative combinations of categorical variables were tested to determine if other factors could explain the large values observed, but none reproduced this pattern. Manual data verification confirmed that all *D*-prefixed *ID* values and *TX operator* occurrences coincided exclusively within the complete-data period from 31/3/2018 to 20/6/2019 as seen on **Visualisation 03**. This dual evidence establishes that the *TX operator* contributes to both the extreme values in Group A variables and the completeness pattern observed across all variables except *sensor6*. This specific finding warrants further investigation into the underlying cause.

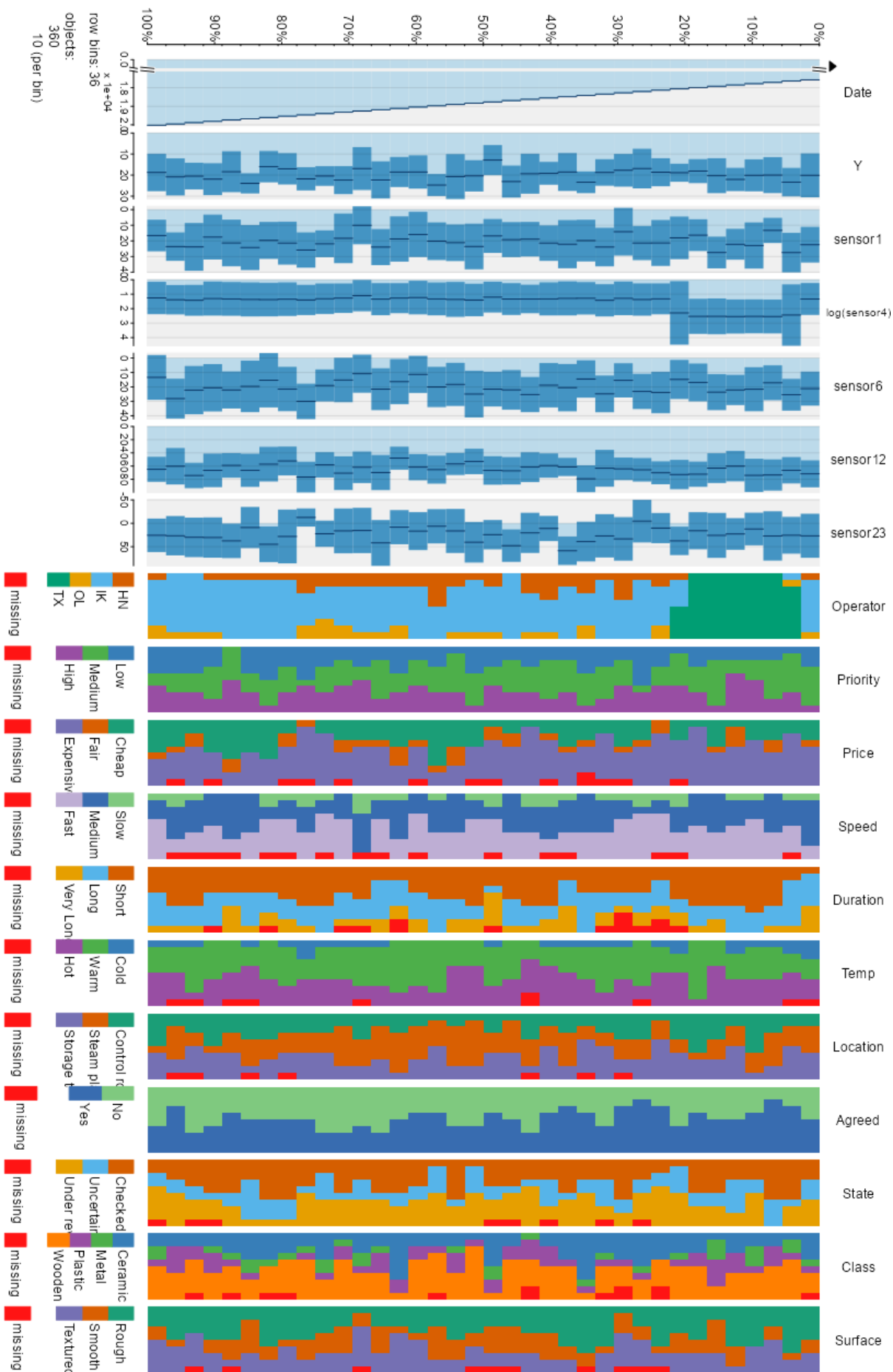


Visualisation 29: Time series plot of Group A sensors with *TX operator* excluded reveals a distinct gap during the period previously containing extreme high values (31/3/2018 to 20/6/2019). Manual verification confirmed all *D*-prefixed *ID* values and *TX operator* occurrences coincided within this complete-data period. Alternative categorical variable combinations were tested but none reproduced this pattern, confirming *TX operator* was uniquely responsible for both the elevated Group A readings and the completeness pattern across all variables except *sensor6*.

11). Tabplot Chart:

A tabplot was used to explore how the categorical variables relate to the numerical variables present in the dataset. This type of visualisation allows multiple categorical dimensions to be viewed simultaneously, while also showing how a numerical variable behaves within each category combination. The size of each segment represents how frequently observations occur, and the colour gradient reflects the average value of the selected numerical variable. As a result, tabplots are useful for identifying concentration effects, potential associations, and differences in numerical behaviour across groups that may not be obvious from standard plots or summary tables.

In this analysis, a tabplot visualisation (**Visualisation 30**) was constructed with time ordering from oldest at the top to newest at the bottom to investigate the relationship between all categorical variables and several key numerical variables, including *Y*, *sensor1*, *sensor4*, *senor6*, *sensor12*, and *sensor23*. To enable fitting within the plot, *sensor4* required log-transformation due to its large range of values, while all remaining numerical variables were left in their original scale. The visualisation highlights a noticeable concentration of observations associated with the *TX* operator during the period from 31/3/2018 to 20/6/2019, further supporting patterns identified in earlier exploratory analysis. Beyond this, no strong or consistent patterns were observed across the other categorical groupings, suggesting that the numerical behaviour of the numeric variables remains relatively stable across all other categories.



Visualisation 30: A tabplot visualisation ordered chronologically (oldest at top, newest at bottom) examines relationships between all categorical variables and key numerical variables (*Y*, *sensor1*, *sensor4*, *sensor6*, *sensor12*, and *sensor23*). *Sensor4* was log-transformed to accommodate its wide range, while all other numerical variables remained untransformed. There are 72 bins with 5 items found across each bin. The plot reveals a notable concentration of TX operator observations during the 31/3/2018 to 20/6/2019 period, consistent with patterns identified in earlier analyses. No other strong associations or trends were detected across remaining categorical groupings, indicating relative stability in numerical variable behaviour across other categories.

12). Mosaic Chart:

Mosaic plots display the proportional relationships between categorical variables. The size of each tile represents frequency, while the colour shading reflects Pearson residuals, indicating whether a combination occurs more or less often than expected under independence. Darker blue shading suggests a higher-than-expected frequency, while red shading suggests lower-than-expected frequency. Statistical significance can be assessed using the reported chi-square p-values.

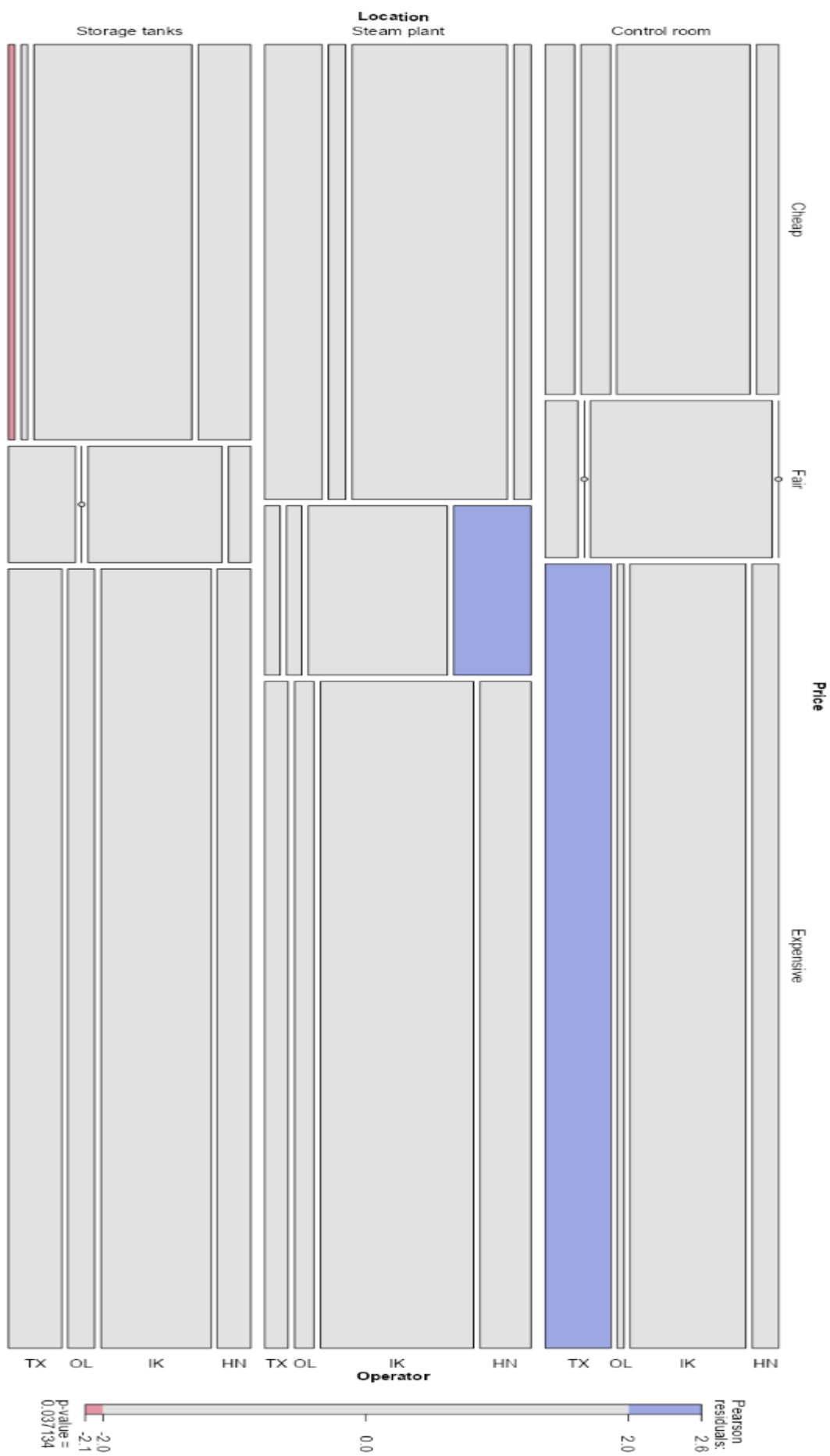
When examining the three-variable mosaic plot of *Location*, *Price*, and *Operator*, there appears to be a statistically significant association between these variables (χ^2 test $p = 0.037$ and statistical significance level of 0.05). Operator *TX* appears to be under-represented within the *Cheap* pricing category at *Storage Tank* locations, as indicated by negative Pearson residual shading. In contrast, *TX* is over-represented in the *Expensive* category within *Control Room* locations, where the darker blue shading suggests that this combination occurs more frequently than expected under independence. Similarly, operator *HN* appears to be over-represented in the *Fair* pricing category at *Steam Plant* locations, reflecting another localised deviation from expected frequencies. Some of these deviations are relatively strong, with Pearson residual values exceedingly approximately ± 2.5 , indicating meaningful separation from independence. This can be seen using **Visualisation 31**.

Beyond the three-way association observed in **Visualisation 31**, examination of all remaining two-, three-, and four-variable combinations of the categorical variables did not reveal statistically significant relationships. In these additional mosaic plots, the associated chi-square tests produced p-values exceeding the conventional 0.05 significance threshold, suggesting insufficient evidence to conclude the presence of meaningful associations. Even though some visual deviations were evident within certain category combinations, these patterns were generally inconsistent and lacked strong statistical support. Because of this, the only categorical interaction demonstrating clear evidence of association was the three-way relationship between *Location*, *Operator*, and *Price*.

13). Data Table:

The original data table was used extensively to perform manual checking of each record, ensuring that the values of a given observation or variable aligned with what each visualisation displayed. This verification process confirmed the accuracy of the plotted data and validated the patterns observed across all visualisations.

Mosaic Plot: Location vs Price vs Operator



Visualisation 31: The mosaic plot reveals distinct association patterns between *Location*, *Price*, and *Operator*. The TX operator shows significant under-representation in Cheap-priced *Storage tank* locations, with observed frequencies falling below expected values under independence. Conversely, TX is markedly over-expressed in Expensive-*Control room* locations, indicating a strong positive association between this operator and high-price, control room combinations. The HN operator also displays notable over-representation in Fair-priced *Steam plant* locations, occurring more frequently than would be expected by chance. These patterns suggest that operator identity is systematically related to both location type and price classification, with TX going towards the most extreme price categories.

14). Conclusion:

Overall, this analysis provided a valuable opportunity to conduct exploratory data analysis on the *Ass1Data* dataset, which resulted in a variety of key findings that can inform further analytical or modelling work.

The investigation revealed distinct sensor groupings that remained consistent across multiple visualisation techniques. Variable *Sensor6* was identified as a notable anomaly with substantially higher missingness at 28.9%. A distinct period of near complete data was also observed between March 2018 and June 2019.

A significant finding emerged regarding the *TX* operator, which was identified as the primary driver behind the extreme values observed in Group A sensors. Temporal analysis confirmed that all *TX* observations occurred exclusively within the complete data period, which established a clear link between operator, data completeness, and extreme value patterns.

Categorical analysis revealed a statistically significant three-way association between *Location*, *Price*, and *Operator* with a p-value approximately 0.037. Notable deviations included *TX* being underrepresented in *Cheap* pricing at *Storage Tank* locations and overrepresented in *Expensive* pricing at *Control Room* locations, while operator *HN* was overrepresented in *Fair* pricing at *Steam Plant* locations. No other categorical combinations produced statistically significant results, indicating this relationship represents the only structured categorical association within the dataset.

The response variable *Y* demonstrated at best moderate associations with sensor variables, with correlations around $r = 0.4$ for sensors in Group E. This suggests that additional contextual information may be required for effective prediction. Future modelling efforts should consider the *TX* operator's distinctive role, consider using the identified sensor clusters as grouped predictors in future modelling work, and perform further exploration to the interaction between *Location*, *Price*, and *Operator*.

15). Use of AI:

During the preparation of this report, DeepSeek and ChatGPT were employed to support code development, identify and resolve coding errors, and refine the written expression of the technical content. These tools were used solely for debugging assistance and stylistic improvements. The underlying data analysis, interpretation of results, and all substantive conclusions are entirely my own.